# Acoustic impedance Z = rho v: the ratio that governs coupling

Acoustic impedance Z = $\rho$ v, impedance table, coupling and matching algebra.

## Acoustic impedance Z = rho v: the ratio that governs coupling

For a travelling plane wave the pressure and the particle velocity $u = -\partial s/\partial t$ are locked in the ratio $\mathbf{Z = \frac{\Delta P}{u} = \rho v}$, the characteristic acoustic impedance (SI unit: rayl = kg m $^{-2}$ s $^{-1}$; 1 rayl $= 1$ Pa s/m). Air at 20 C: $Z = 1.2 \times 343 = 412$ rayl (often rounded to 400); water: $10^3 \times 1483 = 1.48 \times 10^6$ rayl; steel: $7800 \times 5064 = 3.95 \times 10^7$ rayl. The huge air-water mismatch (factor 3600) is why a swimmer hears almost nothing from an airborne source and why the ear needs a middle-ear lever.

## Acoustic impedance Z = rho v: the ratio that governs coupling (con.

Impedance is the acoustic analogue of electrical impedance: pressure plays voltage, particle velocity plays current. Power delivered to a surface: $I = \frac{1}{2}\Delta P_0 u_0 = \frac{1}{2} Z u_0^2 = \frac{\Delta P_0^2}{2Z}$ – same intensity formulas as JEE-10 with $Z = \rho v$ substituted. Sources are rated by impedance matching:

a loudspeaker cone must drive air, so its mechanical impedance should be near $Z_{\text{air}} A$ over its area; the horn shape (Oly 09) is an impedance transformer.

## Acoustic impedance Z = rho v: the ratio that governs coupling (con.

Standing waves: at a displacement node $u = 0$ and $\Delta P \neq 0$, so the local ratio $\Delta P/u$ diverges – the standing wave has position-dependent, purely REACTIVE impedance (like a lossless LC line); only travelling waves carry the real impedance $\rho v$. General wave: $Z = \Delta P/u$ complex; travelling: real $\rho v$; evanescent: imaginary.

WHY / impodance packs two independent properties (density and speed) into one coupling constant: what matters at an interface is not $\rho$ or $v$ separately but their product, because pressure and velocity are the two wave variables.

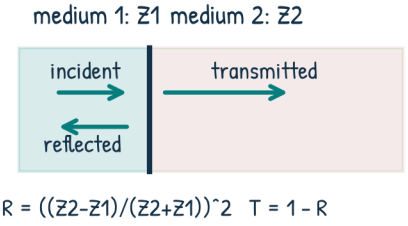
CHECK / units: $\rho v =$ kg m $^{-3}$ m s $^{-1}$ = kg m $^{-2}$ s $^{-1}$ = Pa/(m/s) ok; helium at 20 C: $Z = 0.166 \times 1007 = 167$ rayl – even lower than air despite the high speed, because $\rho$ is tiny: He couples to the ear poorly (soft, quiet) though the speed is high.

## Impedance numbers that decide real problems

Table (rayl): air 412; helium 167; hydrogen 112 ( $0.084 \times 1307$ ... at 20 C $\rho = 0.082$, $v = 1307$: $Z = 107$ ); CO2 500 ( $1.84 \times 272$ ... at 20 C); water $1.48 \times 10^6$; fat tissue $1.38 \times 10^6$ (close to water); bone $7.8 \times 10^6$; steel $3.9 \times 10^7$. Ultrasound imaging: soft-tissue boundaries are weak reflectors (matched); bone and air cavities are strong (mismatched) – the physics of image artefacts.

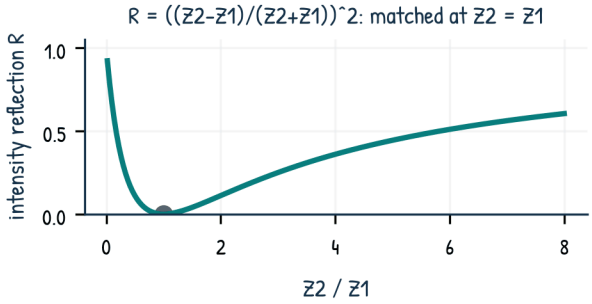
Loudspeaker design: a bare cone radiates into air with efficiency $\sim 2\rho_{\text{air}} v/(\rho_{\text{cone}} v_{\text{cone}})$ -scale mismatch... practical fix: use a large-area light cone (low mechanical $Z$ ) or a horn (Oly 09). The ear solves the reverse problem: air $Z = 412$ to cochlear fluid $Z = 1.5 \times 10^6$:

## interface of two media: impedance mismatch




R = ((Z2-Z1)/(Z2+Z1))^2 T = 1 - R

## reflection fraction vs impedance ratio




REMEMBER / Impedance Z = rho v is the coupling constant; matching Z transfers power, mismatching reflects it.

# Acoustic impedance Z = rho v: the ratio that governs coupling (cont.)

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## Impedance numbers that decide real problems (cont.)

the eardrum area (55 mm$^2$) vs oval window (3.2 mm$^2$) gives an area ratio 17, and the lever of the ossicles another ~1.3: total pressure gain ~22, power transmission improved ~30x; plus the ear canal resonance (JEE-01) adds gain near 3 kHz.

CHECK / reflection fraction for intensity between media $Z_1, Z_2$: $R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2$ (Oly 02): air-water: $R = 0.999$: 99.9% of airborne power bounces off the lake ok why underwater listening to aircraft fails.

specific acoustic impedance vs characteristic: $Z = \rho v$ is per unit area (rayl); mechanical impedance of a panel $Z_{\text{mech}} = \Delta P A / u = ZA$ in ohms-kg/s — the driving-point quantity for motors and diaphragms.

## Impedance algebra drills

(a) Same intensity in air and water: pressure amplitudes scale as $\sqrt{Z}$: $\Delta P_{\text{water}} / \Delta P_{\text{air}} = \sqrt{1.48 \times 10^6 / 412} = 60$; underwater sound of the same $I$ has 60x the pressure swing. (b) $s_0$ at same $I$: $s_0 \propto 1/\sqrt{Z\omega^2}$ -ish: water displacement for same $I$ is $1/60$ of air (times speed ratio... precisely $s_0 = \Delta P_0 / (Z\omega)$: $\Delta P_0$ 60x and $Z$ 3600x: $s_0$ 60x smaller ok). (c) Design: a transducer of area 1 cm$^2$ driving air at 1 Pa amplitude delivers $\Delta P^2 A/(2Z) = 1 \times 10^{-4}/824 = 1.2 \times 10^{-7}$ W — microwatts; the same face in water delivers 0.34 W — water loads transducers heavily.

## Impedance algebra drills (cont.)

Impedance ladder across a problem: media ($Z = \rho v$), then interface factors ($R$, $T$), then geometry (spreading), then absorption: the four multipliers of every acoustic power budget.

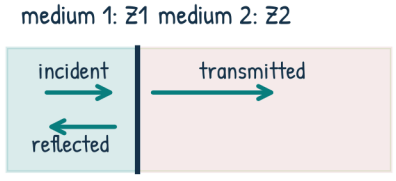
WHY / engineers think in dB of mismatch loss; physicists think in $Z$ ratios; exams give $\rho$ and $v$ and expect the product — never forget the product is the coupling constant.

CHECK / the ear canal and Helmholtz resonances are IMPEDANCE phenomena (reactance cancellation): the resonance formulas of JEE-17/19 reappear as impedance matching — one physics, two vocabularies.

## Impedance view of amplitude relations (consolidation)

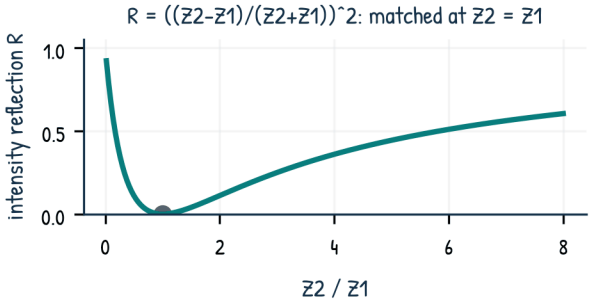
Re-derive the amplitude toolkit through $Z$: $I = \frac{\Delta P_0^2}{2Z} = \frac{1}{2} Z \omega^2 s_0^2$ (since $\Delta P_0 = Z\omega s_0$); $\Delta P_0 = \sqrt{2ZI}$; $s_0 = \frac{\Delta P_0}{Z\omega}$. Intensity level: $\beta = 10 \log_{10} \frac{\Delta P_0^2}{2ZI_0}$ — note the level of a FIXED pressure amplitude depends on the medium! 1 Pa is 113.8 dB in air ($20\log(1/2.87 \times 10^{-5})$) but only 90.7 dB in water (referenced to the same $I_0$): the reference convention matters across media.

### interface of two media: impedance mismatch




R = ((Z2-Z1)/(Z2+Z1))^2  T = 1 - R

### reflection fraction vs impedance ratio




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# Acoustic impedance Z = rho v: the ratio that governs coupling (cont.)

Acoustic impedance Z = $\rho$ v, impedance table, coupling and matching algebra.

## Impedance view of amplitude relations (consolidation) (cont.)

Impedance and Doppler (bridge): a moving source radiates into the medium with Doppler factors; the POWER of the received wave scales as $f'^2$ -ish through the amplitude change (Oly 05):
$I \propto \Delta P_0^2/2Z$ with $\Delta P_0$ Doppler-shifted — the intensity Doppler factor is $\left(\frac{f'}{f}\right)^2$ for moving-source... precisely $I'/I = (f'/f)^3$ for sound (energy flux + frequency); state assumptions carefully.
the specific impedance is real only for progressive waves; measurement instruments quote $Z$ of materials as $\rho v$ with $\pm$ few-% uncertainty from $\rho$ and $v$ errors — propagate independently: $\delta Z/Z = \delta\rho/\rho + \delta v/v$.

## Reflection and transmission at boundaries: R = left(fracZ 2 - Z 1Z.

At a plane interface between media of impedances $Z_1, Z_2$ the pressure reflection coefficient $r_p = \frac{Z_2 - Z_1}{Z_2 + Z_1}$ (signed) and the intensity reflection fraction $R = r_p^2 = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2$;
transmission $T = 1 - R = \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2}$. Extreme cases: $Z_2 \gg Z_1$ (air to wall): $r_p \to +1$ (pressure reflects in phase, hard wall, displacement node in the air); $Z_2 \ll Z_1$ (water to air): $r_p \to -1$ (pressure inverts, soft boundary, pressure node); $Z_2 = Z_1$ : $R = 0$, perfect transmission — the impedance-matching ideal.

## Reflection and transmission at boundaries: R = left(fracZ 2 - Z 1Z.

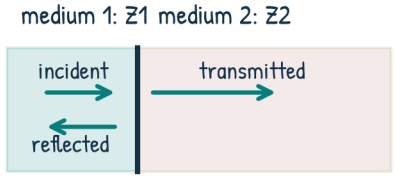
The particle velocity coefficient has the opposite sign pattern: $r_u = -r_p$ : at a hard wall velocity inverts (displacement node ok JEE-16); at a soft end pressure inverts (pressure node ok). Derivation: continuity of pressure and of normal velocity across the interface: $1 + r = t$ and $(1 - r)/Z_1 = t/Z_2$ solve to the boxed pair.
Worked numbers: air ( $Z = 412$ ) to glass ( $Z = 1.3 \times 10^7$ ): $R = 0.99987$ — a glass wall reflects 99.99% of intensity (why rooms ring); air-water: $R = 0.999$ ; water-glass: $R = 0.63$ ... compute: $((1.3e7 - 1.48e6)/(1.3e7 + 1.48e6))^2 = 0.63$ : 63% — soft by comparison; fat-kidney soft tissue ( $Z$ similar): $R < 0.01$ — why ultrasound needs speckle and Doppler contrast.

## Reflection and transmission at boundaries: R = left(fracZ 2 - Z 1Z.

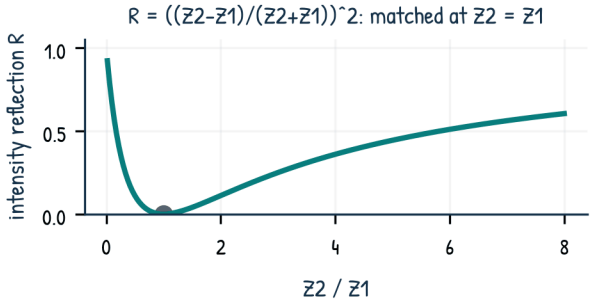
WHY / the boundary conditions are the two continuity laws (pressure continuous — no infinite force; normal velocity continuous — media stay in contact); everything else follows algebraically; the square for intensity comes from $I \propto \Delta P_0^2/Z$.

CHECK / energy bookkeeping: $R + T = 1$ exactly; a computed $R > 1$ or $T > 1$ signals a sign error; $r_p$ sign flips tell the phase of the reflected wave: wall-type (positive, pressure antinode at boundary) vs open-type (negative, pressure node).

## interface of two media: impedance mismatch




## reflection fraction vs impedance ratio




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# Reflection and transmission at boundaries: R = left(fracZ 2 - Z 1Z 2 + Z 1right)2

Boundary reflection/transmission coefficients, quarter-wave matching, phase rules.

## Quarter-wave layers and impedance matching

A layer of thickness $\lambda/4$ and impedance $Z_l = \sqrt{Z_1 Z_2}$ cancels reflection at one frequency: the two reflected waves from the layer faces interfere destructively when $Z_l^2 = Z_1 Z_2$ (anti-reflection coating; the transformer equation $Z_{\text{in}} = Z_l^2 / Z_{\text{load}}$ — acoustic transmission line theory). Exam version:
"what thickness of material between air and wall removes reflection at 1 kHz?" $\lambda/4$ of the layer: $v_l/4f$.

Standing wave ratio as a measurable: $\text{SWR} = \dfrac{1+|r_p|}{1-|r_p|}$; measuring $I_{\max}/I_{\min} = \text{SWR}^2$ in a duct recovers $|r_p|$ and hence $Z_2$ — the standard impedance-measurement method.

## Quarter-wave layers and impedance matching (cont.)

Partial reflection in pipes: an open end radiates; $R = 1 - (ka)^2/2$ -ish for low $ka$ ... precisely: radiation from an unflanged pipe end reflects intensity $R \approx 1 - \frac{1}{2}(ka)^2$ at low frequency (end correction is the reactive part of the same impedance, JEE-19).

WHY / the $\lambda/4$ transformer works because the round trip inside the layer is a half wavelength: the second reflection is $180°$ out of phase with the first when the impedance chain is geometric-mean matched.

## Quarter-wave layers and impedance matching (cont.)

CHECK / coating demo: glass coated so $Z_l = \sqrt{Z_{\text{air}} Z_{\text{glass}}}$ : $Z_l = 2.3 \times 10^5$ rayl — e.g. a porous layer of the right density and speed; quarter-wave at 1 kHz: thickness $v_l/4000$ ... with $v_l = 1000$ m/s: 25 cm — unrealistically thick for air-acoustics at 1 kHz: matching layers are practical in ultrasound (MHz, mm scales) ok.

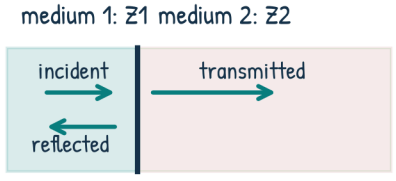
Hard wall ( $Z_2 \gg Z_1$ ): $r_p = +1$ : pressure reflects UNINVERTED (antinode at wall); $r_u = -1$ : velocity inverts (node at wall) ok JEE-16 boundary rules derived, not memorised. Open end ( $Z_2 \ll Z_1$ , e.g. pipe opening to atmosphere): $r_p = -1$ (pressure node just outside), $r_u = +1$ (velocity antinode).

## Phase rules from impedance (the safe derivation) (cont.)

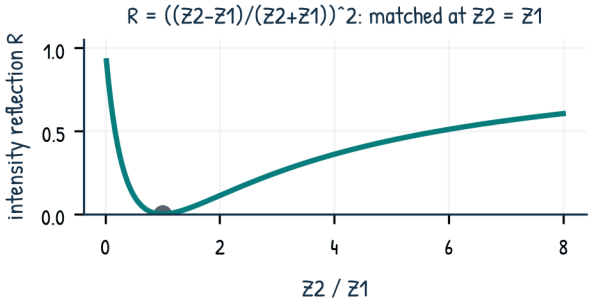
The quarter-wave offset between pressure and displacement in travelling waves (SN-03) reappears here as the $\lambda/4$ displacement-vs-pressure offset of the standing pattern.

Why the wall reflects: air cannot drive the dense wall (huge $Z$ ), so the velocity at the wall is forced to zero: the superposition of incident and reflected waves must cancel velocity there: $u_i + u_r = 0 \to r_u = -1 \to r_p = +1$ ok consistent with the formulas.

incident, reflected, transmitted at a boundary




R = ((Z2-Z1)/(Z2+Z1))^2 curve




REMEMBER / R = ((Z2 - Z1)/(Z2 + Z1))^2, T = 4 Z1 Z2/(Z1 + Z2)^2; hard walls keep pressure, soft ends kill it.

# Reflection and transmission at boundaries: R = left(fracZ 2 - Z 1Z 2 + Z 1right)2 (cont.)

Boundary reflection/transmission coefficients, quarter-wave matching, phase rules.

## Phase rules from impedance (the safe derivation) (cont.)

Soft-end subtlety: the open end is NOT exactly $Z = 0$: the radiation impedance $Z_{\text{end}} = \rho v[\frac{1}{4}(ka)^2 + i \cdot 0.6ka]$ -ish (Oly 05): real part radiates (why $R < 1$ slightly), imaginary part is the end correction $e = 0.6a$ (JEE-19) — the three topics (reflection, end correction, radiation resistance) are one impedance expansion.

CHECK / plug $Z_2/Z_1 = 10^5$: $r_p = 0.99998$: $R = 0.99996$; $r_u = -0.99998$ ok hard; $Z_2/Z_1 = 10^{-5}$: $r_p = -1$: soft ok.

## Reflection problem patterns

(a) Ultrasound pulse between soft tissues: 2% reflection at a boundary of $R = 0.01$: how many echoes reach the probe from 3 interfaces? cascade $T^2R$ products: $0.99^2 \times 0.01$ -scale each — the brightness law of ultrasound. (b) "Find $Z_2$ if $R = 0.25$ ": $|r_p| = 0.5$: $Z_2/Z_1 = 3$ or 1/3. (c) A tube closed by a movable wall of impedance $Z_2$:

## Reflection problem patterns (cont.)

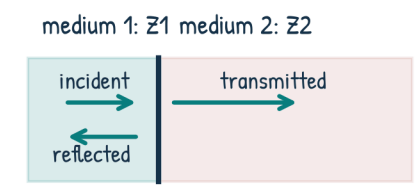
resonance conditions shift from pure node/antinode (JEE-17) toward partial standing waves: the boundary displacement is finite: acoustic load problem.

## Reflection problem patterns (cont.)

WATCH / Traps: (i) $R$ vs $r_p$: intensity uses the SQUARE; pressure amplitude uses $|r_p|$; (ii) phase rules: state which variable (P or u) inverts; (iii) $R + T = 1$ only for INTENSITY at a lossless boundary — with absorption $R + T + A = 1$.

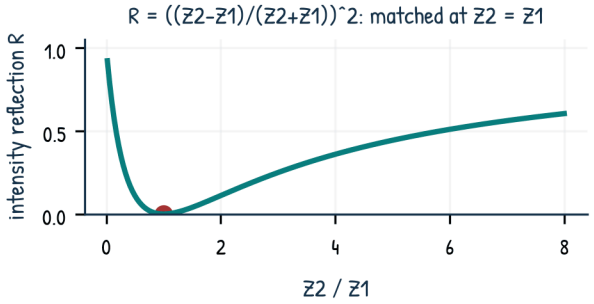
WHY / the same few numbers ($Z$ ratios) decide hearing, room acoustics, ultrasound imaging, and sonar targets — one boundary table worth memorising: $R$, $r_p$ sign, wall type.

incident, reflected, transmitted at a boundary




R = ((Z2-Z1)/(Z2+Z1))^2  T = 1 - R

R = ((Z2-Z1)/(Z2+Z1))^2 curve




REMEMBER / R = ((Z2 - Z1)/(Z2 + Z1))^2, T = 4 Z1 Z2/(Z1 + Z2)^2; hard walls keep pressure, soft ends kill it.

# Reflection and transmission at boundaries: R = left(fracZ 2 - Z 1Z 2 + Z 1right)2 (cont.)

Boundary reflection/transmission coefficients, quarter-wave matching, phase rules.

## Reflection problem patterns (cont.)

CHECK / special values: $Z_2 = 3Z_1$ : $R = (2/4)^2 = 0.25$ , $T = 0.75$ ok; $Z_2 = Z_1/3$ : $R = 0.25$ ok symmetric in log: the reflection is the same whether the wave enters the denser or rarer side (intensity), only the pressure phase differs.

A pipe of length $L$ driven at one end presents an input impedance $Z_{\text{in}}$ that depends on the far-end load. Lossless line theory: $Z_{\text{in}} = Z_0 \dfrac{Z_L + iZ_0 \tan kL}{Z_0 + iZ_L \tan kL}$ with $Z_0 = \rho v/S$ the characteristic line impedance (per area $S$ ), $Z_L$ the load. Resonance (maximum amplitude for a given drive) occurs when $Z_{\text{in}}$ is minimum (open-open:

## Pipes as impedance problems: input impedance and resonance (cont.

$Z_L = 0$ : $Z_{\text{in}} = iZ_0 \tan kL = 0$ at $kL = n\pi \rightarrow$ $L = n\lambda/2$ ok all harmonics; closed-open: $Z_L = \infty$ : $Z_{\text{in}} = -iZ_0 \cot kL = 0$ at $kL = (2n-1)\pi/2 \rightarrow$ odd quarter waves ok JEE-17/18 derived from one formula).

Why closed pipes give odd harmonics: the mismatched ends force $\tan kL$ and $\cot kL$ conditions alternating: counting quarter wavelengths that fit between a velocity node and a pressure node is the SAME as setting the imaginary part of $Z_{\text{in}}$ to zero. Mixed-end resonance spacing $\Delta f = v/2L$ falls out of the $\tan$ period π in $kL$ .

## Pipes as impedance problems: input impedance and resonance (cont.

Load effects: a pipe with a partially open far end ( $Z_L$ finite, radiation) has damped resonance peaks and shifted frequencies — the Q factor (Oly 05) comes from the real part of $Z_{\text{in}}$ ; the end correction (JEE-19) is the imaginary part at the mouth; a flange changes $Z_L$ and thus $e$ (0.6r to 0.85r) consistently.

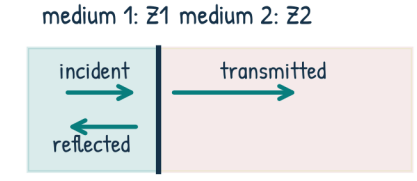
WHY / impedance language unifies: boundary conditions, end corrections, resonance ladders, radiation, and damping are all one rational function $Z_{\text{in}}(kL, Z_L)$ — the exam formulas are its special cases.

## Pipes as impedance problems: input impedance and resonance (cont.

CHECK / specialise: $L = \lambda/2$ open-open: $kL = \pi$ : $\tan = 0$ : $Z_{\text{in}} = 0$ ok short at the drive point: maximum flow for given pressure... resonance of a driven pipe is when the drive sees minimum impedance ok; closed-closed: $Z_L = \infty$ both ends: $kL = n\pi$ again all harmonics ok JEE-16 CHECK.

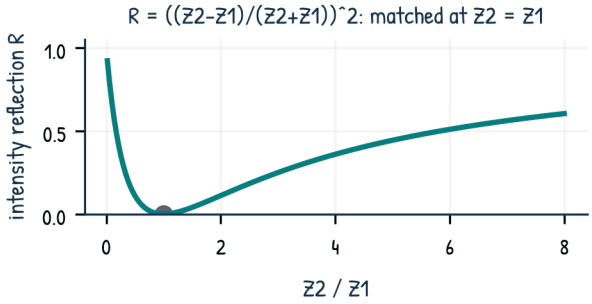
Driven pipe amplitude vs frequency: Lorentzian of half-width $\Delta f = f_1/Q$ : $Q = \dfrac{\pi}{2\ln(1/r)}$ -ish for end reflection $r$ ... working rule: $Q \approx \dfrac{\pi}{2(1-r)}$ for $r \rightarrow 1$ (small losses); typical unflanged pipe $Q \sim 10$ -30; flanged 20-60; with damping wool less.

### incident, reflected, transmitted at a boundary




R = ((Z2-Z1)/(Z2+Z1))^2  T = 1 - R

### R = ((Z2-Z1)/(Z2+Z1))^2 curve




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# Pipes as impedance problems: input impedance and resonance

Input impedance of pipes, resonance/Q, why odd harmonics for closed pipes.

## Resonance curve: width, Q, and the amplitude build-up (cont.)

At resonance the amplitude gain over the non-resonant case is $\sim Q$ (in pressure); bandwidth $\Delta f$ measures the fractional loss per cycle.

Time picture: a driven resonator rings up over $\tau = Q/(\pi f_1)$ ... precisely $\tau = \dfrac{Q}{\pi f_1} = \dfrac{1}{\pi \Delta f}$ (energy decay time $\tau_E = Q/(2\pi f)$ -ish); striking a pipe: the sound decays exponentially with that time constant: exam estimate: Q = 20 at 300 Hz: $\tau = 21$ ms.

Multiple coupled pipes (organ ranks, bottles): coupling shifts each resonance slightly (mode splitting); a Helmholtz resonator on a wall acts as a notch absorber at $f_H = \dfrac{v}{2\pi}\sqrt{S/(V L_{\text{eff}})}$ — room acoustics tool built on pipe theory.

## Resonance curve: width, Q, and the amplitude build-up (cont.)

CHECK / resonance width example: $f_1 = 300$ Hz, $Q = 15$: half-power width 20 Hz: the resonance tube (JEE-20) length sweep through a peak of that width in frequency terms: $\Delta L = \lambda \Delta f/(2f) = 2$ cm for the second resonance spacing... order: noticeable in careful labs, negligible in rough ones.

WHY / Q is a number for how many cycles energy survives: $Q = 2\pi \times$ energy stored / energy lost per cycle — the single number that decides whether a resonance is sharp (tuning forks Q ~ 1000, pipes Q ~ 20, rooms Q ~ 5).

## Impedance methods in exam clothing

(a) "Why does a closed pipe sound hollow?" odd harmonics only = the $\cot kL$ resonance sequence; explain via impedance: the closed end is a velocity node (infinite load), the open end a pressure node (zero load): only odd quarters fit. (b) "Predict the effect of a movable piston in a tube": $L$ varies: resonance ladder sweeps; with $Z_L$ finite at the drive, peaks broaden.

(c) "Design a tube resonant at 440 Hz and 660 Hz": open-open with $f_1 = 220$: $L = v/2f_1 = 0.78$ m (minus 2e).

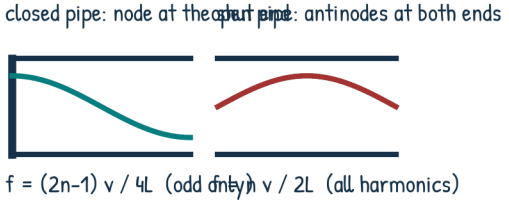
Half-open side branch (quarter-wave trap): a branch of length $\lambda/4$ closed at the far end presents $Z_{\text{in}} = 0$ at the junction (short circuit for waves of that $f$): mufflers use arrays of such traps — mechanical filter design, exam-explainable.

## Impedance methods in exam clothing (cont.)

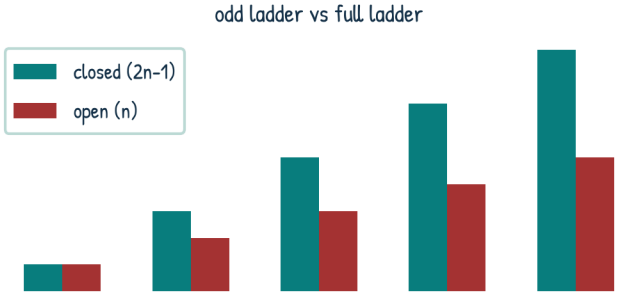
the general line equation: $\dfrac{\partial^2 P}{\partial x^2} = \dfrac{1}{c^2}\dfrac{\partial^2 P}{\partial t^2}$ for the pressure in a duct of constant area (same wave equation, 1D): the impedance formalism is the travelling/standing decomposition of its solutions.

(a) Open pipe $L = 0.4$ m, $r = 2$ cm: $L_{\text{eff}} = 0.424$ m: $f_1 = 343/0.848 = 404.5$ Hz; $Q = 20$: bandwidth 20 Hz; driving at 395 Hz: amplitude $\sim Q/2$ -ish below peak... half-power at $\Delta f = \pm 10$ Hz: 395 Hz is 9.5 Hz off: $\sim 1/\sqrt{2}$ of peak amplitude. (b) Closed pipe same length: $f_1 = 343/(4 \times 0.412) = 208$ Hz (one end corrected only).

### pipe as a transmission line with a load




### resonance peak width set by Q




REMEMBER / A pipe is a load on a line: resonance is zero input reactance; closed pipes give odd harmonics by cot(kL).

# Pipes as impedance problems: input impedance and resonance (cont.)

Input impedance of pipes, resonance/Q, why odd harmonics for closed pipes.

## Resonance numbers drill (impedance view) (cont.)

End-loaded pipe: with $Z_L = Z_0/2$ (e.g. a partially open damper): resonance shifts between the open and closed ladders; solving $\tan kL = Z_0/Z_L$ ... the transcendental decides: exam problems state "closed" or "open" only.

WHY / the impedance method gives answers without redrawing node pictures: write $Z_{\text{in}}$, set the reactance to zero, read the ladder — faster and safer under time pressure.

A harmonic wave at a point is a rotating phasor of length $a$ and phase $\phi$: $s = a\cos(\omega t - \phi) \leftrightarrow ae^{i\phi}$. Sum of many same-frequency waves = phasor polygon: $A = \left|\sum_k a_k e^{i\phi_k}\right|$; phase $\arg\left(\sum\right)$. Two phasors:

## Phasors: superposition of harmonic waves as vector addition (cont.)

$A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\Delta\phi}$ (the JEE-13 formula, now a picture); equal amplitudes: $A = 2a\cos(\Delta\phi/2)$ bisecting the angle.

Why phasors work: the wave equation is linear and time-harmonic; the time factor $e^{-i\omega t}$ cancels in all ratios and sums; only spatial phases matter. Interference problems become geometry: equal spacing $\delta = \lambda/N$ between $N$ sources closes the polygon (partial cancellation); symmetric arrangements give closed polygons (silence: e.g.

3 sources at the vertices of an equilateral triangle seen from the centre: phases 0, 120, 240: sum zero).

## Phasors: superposition of harmonic waves as vector addition (cont.)

Beats revisited: two phasors of nearby frequency rotate at slightly different rates: the resultant length oscillates at the difference rate — the beat envelope IS the rotating-vector picture (JEE-15); the phasor tip traces an epicycloid.

WHY / complex exponentials turn differentiation into multiplication ($\partial_t \to -i\omega$): the wave equation becomes algebraic; every resonance, impedance, and interference formula of this chapter is a statement about complex sums.

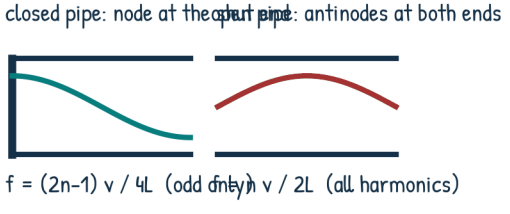
CHECK / three-source check: $a, a, a$ with phases 0, 120°, 240°: sum = 0 ok (used in three-phase cancellation and in $N$-slit minima conditions $\delta = \lambda/N$ multiples... precisely minima of $N$-source interference when $N\delta/\lambda$ is a non-multiple-of-N integer).

## Phasor clinic: worked sums

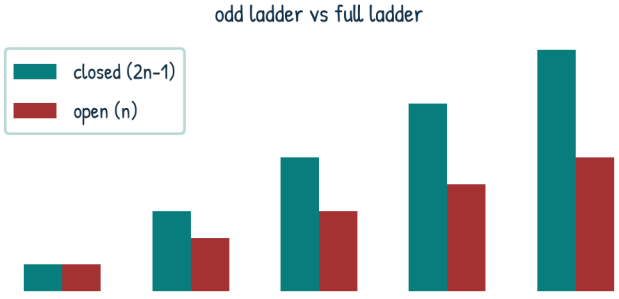
(a) $a_1 = 3$, $a_2 = 4$, $\Delta\phi = 90°$: $A = 5$; phase $\tan^{-1}(4/3) = 53.1°$ from phasor 1. (b) $N = 4$ equal phasors with successive phase $\delta = 60°$: closed-chain geometry: $A = a\dfrac{\sin(N\delta/2)}{\sin(\delta/2)} = a\dfrac{\sin 120°}{\sin 30°} = 1.73a$ — the array-factor formula (same as $N$-slit optics).

(c) Two waves of amplitude ratio 3:1 in antiphase: $A = 2a$; the "zero" of interference is incomplete (contrast $\mathcal{V} = 2\cdot 3/(9+1) = 0.6$).

## pipe as a transmission line with a load




## resonance peak width set by Q




REMEMBER / A pipe is a load on a line: resonance is zero input reactance; closed pipes give odd harmonics by cot(kL).

# Phasors: superposition of harmonic waves as vector addition

Phasor addition, array factor sin(N $\psi$ /2)/sin( $\psi$ /2), quadrature as rotation.

## Phasor clinic: worked sums (cont.)

Array factor: $A(\theta) = a_0 \frac{\sin(N\psi/2)}{\sin(\psi/2)}$ with $\psi = kd\cos\theta + \beta$ the progressive phase — loudspeaker arrays and diffraction gratings in one formula; principal maxima at $\psi = 2\pi m$.

Power of the picture: instead of trig identities, DRAW: equal-length chords, inscribed polygon, radius $R = a/2\sin(\psi/2)$ -ish; the sum is the closing chord — Euclidean geometry replaces algebra.

CHECK / array-factor sanity: $\psi = 0$ : $A = Na$ ok (the $\sin/\sin$ limit); $\psi = 2\pi/N$ : $A = 0$ ? $\sin(N \cdot \pi/N) = \sin\pi = 0$ ok first minimum; for $N = 2$ : $A = 2a\cos(\psi/2)$ ok beats/interference formula recovered.

## Phasors for standing waves and quadrature energy

Standing wave as two counter-rotating phasors (equal $\omega$, opposite direction): at a node they cancel always (antiparallel); at an antinode they add. The pressure-displacement quadrature (SN-03) is a $90^\circ$ phasor offset: energy shuttles between the two quadrature components, total constant for a standing wave.

Energy in phasor language: for $s = a\cos(\omega t - \phi)$, $u = \partial_t s = a\omega\sin(\omega t - \phi)$ — velocity phasor leads displacement by $90^\circ$; the impedance $Z = \Delta P/u$ is the ratio of two phasors; reactive impedance = $90^\circ$ offset (stored energy), real = in-phase (radiated power).

Power factor of a source: a driver in phase with the velocity radiates max power (real $Z$ ); $90^\circ$ out of phase it stores (near-field reactance) — antenna and loudspeaker design language.

## Phasors for standing waves and quadrature energy (cont.)

WHY / the phasor at $90^\circ$ is the mathematical seat of the quadrature physics: every "X leads Y by 90°" statement in this chapter is a phasor sentence.

CHECK / round trip: $\Delta P = -B\partial s/\partial x$ is a spatial derivative = $90^\circ$ phase rotation for sinusoids: pressure phasor = displacement phasor rotated by $-90^\circ$ and scaled by $Bk = Z\omega$ ok consistent with $\Delta P_0 = Z\omega s_0$.

(a) Two speakers driven in phase, listener at path difference $3\lambda/4$ : $\Delta\phi = 270^\circ$ : amplitude $2a|\cos 135^\circ| = 0.71 \cdot 2a$ ... $\sqrt{2}\,a$ of the sum-free... precisely $2a\cos(135^\circ) = -\sqrt{2}\,a$ : amplitude $\sqrt{2}\,a$ : intensity 2x one source. (b) Three equal sources, phases 0, 90°, 180°: $A = a|1 + i - 1| = a$ :

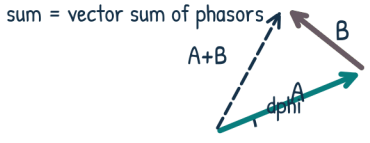
## Phasor drill set (cont.)

intensity = one source. (c) Balanced bridge of four speakers in a square, driven alternately in phase/antiphase: at the centre all pairs cancel — architectural silence.

General N: geometric series $\sum_{k=0}^{N-1} ae^{ik\psi} = a\frac{1 - e^{iN\psi}}{1 - e^{i\psi}}$ — derive the array factor in one line; the phase-only version drives phased-array sonar.

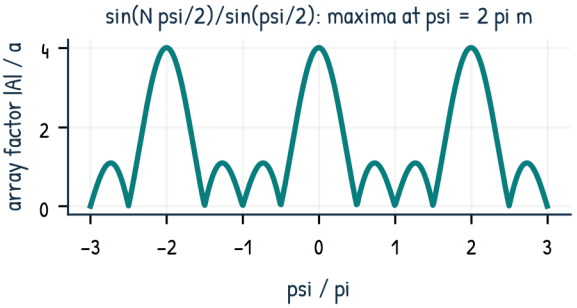
phasors extend to noise and pulses by Fourier decomposition: any waveform = phasor sum; linear systems act on each phasor independently with a transfer function $H(\omega)$ — the door to signal processing.

### phasor addition: A + B as vectors




|A+B|^2 = A^2 + B^2 + 2 A B cos dphi

### array factor of N coherent sources




REMEMBER / Phasors turn interference into vector sums; the array factor sin(N psi/2)/sin(psi/2) covers N sources.

# Energy flux, radiation resistance, and the Q of resonance

Radiation resistance, near/far field, Q = $\omega$ E/P, ring-down and bandwidth.

## Energy flux, radiation resistance, and the Q of resonance

Time-averaged intensity of a travelling wave: $I = \bar{u}v = \dfrac{\Delta P_0^2}{2Z}$ with $Z = \rho v$ (Oly 01). For a general (possibly reactive) field the instantaneous flux $\Pi = \Delta P \cdot u$ oscillates; only its average transports. A driven resonator near resonance stores energy $\mathcal{E}$ and loses power $P_{\text{loss}} = \omega\mathcal{E}/Q$;

quality factor $Q = \dfrac{\omega\mathcal{E}}{P_{\text{loss}}} = 2\pi\dfrac{\mathcal{E}}{\text{loss per cycle}}$;

ring-down time $\tau = Q/\omega$ (amplitude) — the number that separates tuning forks ( $Q \sim 10^3$ ) from rooms ( $Q \sim 5$ ).

## Energy flux, radiation resistance, and the Q of resonance (cont.)

Radiation resistance: a vibrating surface of area $S$ and velocity amplitude $u_0$ radiates $P = \frac{1}{2}R_r S u_0^2$ with $R_r \approx Z_{\text{air}}$ for a large baffle ( $R_r = Z$ per unit area) but $R_r \propto (ka)^2 Z$ for a small piston ( $ka < 1$ ): small sources are BAD radiators — the reason a bare vibrating rod is quiet and a loudspeaker cone is big. Multipole order: monopole (pulsating sphere) radiates best at low $ka$; dipole (oscillating sphere) loses another $(ka)^2$.

Near field vs far field: within $r < \lambda/2\pi$ the reactive component dominates ( $\Delta P \perp u$ ): energy sloshes, no net flux; beyond, the travelling wave $\Delta P = Zu$ carries $I \propto 1/r^2$. Microphone placement and speaker measurements must sit in the far field.

## Energy flux, radiation resistance, and the Q of resonance (cont.)

WHY / Q is the single number connecting ring-down, bandwidth, and amplitude build-up: $\Delta f = f/Q$, $\tau = Q/(2\pi f)$ -ish in time units, and peak gain $\propto Q$ for a given drive — one concept, three exam formulas.

CHECK / consistency: $Q = 2\pi\mathcal{E}/W_{\text{cycle}}$; $\Delta f = f_1/Q$; $\tau_{\text{amp}} = Q/(\pi f_1)$ ... at $Q = 20$, $f = 300$ Hz: $\tau = 21$ ms and $\Delta f = 15$ Hz: a struck pipe fades in ~3 ringing times ≈ 60 ms ok audible clap-coloration.

Pulsating sphere radius $a$, $ka < 1$: $R_r = \rho c (ka)^2/4$ per area... source power $P = 2\pi a^2 u_0^2 \rho c (ka)^2/4$ ...

exam form: $P = \dfrac{\rho c k^2 u_0^2}{4\pi} V^2$ -type scaling with volume velocity $Q_v = u_0 S$; a 5 cm speaker at 1 kHz ( $ka = 0.9$ ): roughly $R_r \sim Z/2$: efficient;

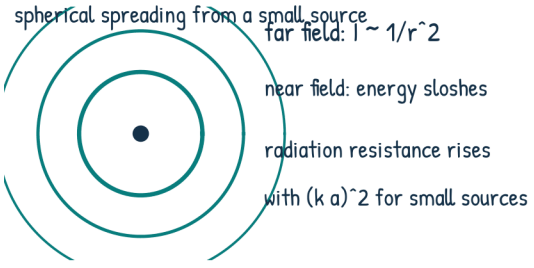
## Radiation from real sources (monopole, dipole, baffles) (cont.)

the same at 100 Hz ( $ka = 0.09$ ): $R_r \sim Z/250$: wimpy bass — why woofers are large and ported.

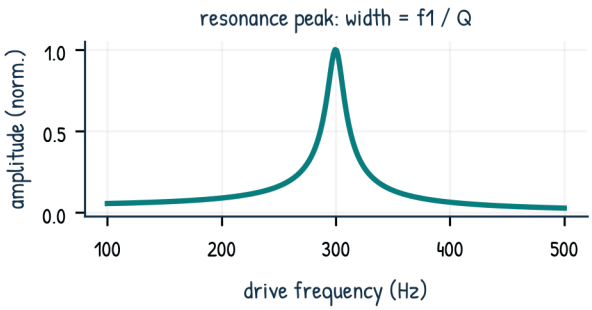
Baffle and enclosure effects: an unbaffled driver short-circuits front-to-back at low $ka$ (dipole cancellation, 6 dB/octave rolloff); a box or horn restores monopole radiation; the infinite baffle gives $R_r = Z$ per area (pressure doubling at the panel).

Biological speakers: the human voice ~1 cm $^2$ glottis at 100-300 Hz: $ka < 0.1$: extremely inefficient (~0.01% of metabolic power): the mouth and vocal tract resonance (formants) do the impedance matching; birds sing 2-8 kHz where $ka \sim 1$: efficient little radiators.

## spherical spreading and the radiating surface




## resonance curve: width f1/Q




REMEMBER / Q = omega E / P_loss ties ring-down, bandwidth and gain; small sources are poor radiators.

# Energy flux, radiation resistance, and the Q of resonance (cont.)

Radiation resistance, near/far field, Q = $\omega$ E/P, ring-down and bandwidth.

## Radiation from real sources (monopole, dipole, baffles) (cont.)

CHECK / power scaling: $P \propto u_0^2 f^2$ at low $ka$ for a fixed source: doubling frequency from 100 to 200 Hz multiplies radiated power by 4 at the same diaphragm speed ok why tweeters are easy and woofers hard.

(a) A 0.5 W source at 340 Hz in air: $I = P/(4\pi r^2)$: at 5 m: $1.6 \times 10^{-3}$ W/m$^2$ = 92 dB; $\Delta P_0 = \sqrt{2ZI} = 1.14$ Pa; $s_0 = \Delta P_0/(Z\omega) = 1.3\mu$ m. (b) "How much power to hold 1 Pa at 1 kHz over 1 m$^2$?" $P = \Delta P_0^2 S/(2Z) = 1.2$ W — modest; for 100 Pa (pain) 12 kW over 1 m$^2$ — concert-scale.

## Energy-budget problems (cont.)

(c) A fork of Q = 500, $f = 256$ Hz struck to 1 mm amplitude: stored energy $\mathcal{E} \sim \frac{1}{2} m_{\text{eff}} \omega^2 x^2$: for $m_{\text{eff}} = 20$ g: $1.3 \times 10^{-3}$ J; loss per cycle $\mathcal{E} \cdot 2\pi/Q = 16\,\mu$J; ring-down $\tau = Q/(\pi f) = 0.62$ s.

Acoustic vs perceived: equal-loudness (Fletcher-Munson): physical $I$ must rise at low $f$ to match perceived loudness — power budget of hi-fi at 40 Hz is 10–20 dB more than at 1 kHz.

WHY / energy methods shortcut amplitude problems: instead of solving the wave equation, balance $P_{\text{source}} = \oint I\,dA + d\mathcal{E}/dt + P_{\text{loss}}$ — the acoustic version of the work-energy theorem.

## Q and resonance in the exam frame

(a) "A closed pipe Q = 12 at f1 = 200 Hz is driven: amplitude at resonance vs 10 Hz off": ratio $\approx Q\Delta f/f$ -ish: at $\delta = 10$ Hz = half-width: amplitude $1/\sqrt{2}$ of peak.

(b) "Two pipes of equal f1 but Q = 10 and 40: which is louder at resonance, which more selective?" same peak if loss-limited... high-Q rings louder after excitation and narrower in sweep.

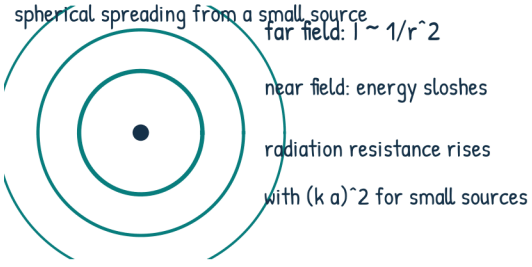
(c) "Estimate the Q of a room from reverberation": $Q = \pi f T_{60}/6.9$ ... standard: $T_{60} = 0.161 V/A$ (Sabine): Q from bandwidth of a mode: $Q = fT_{60}/2.2$ -ish: at 100 Hz, $T_{60} = 0.5$ s: $Q = 23$.

## Q and resonance in the exam frame (cont.)

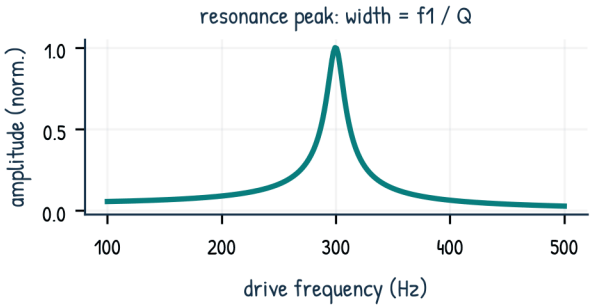
WATCH / Traps: (i) $\tau = Q/\omega$ vs $Q/(\pi f)$: factor conventions — state which amplitude time you use; (ii) Q is defined for ENERGY; amplitude decays twice as slowly... amplitude time $\tau_A = 2\tau_E$: exam convention varies: give both. (iii) Nonlinear large-amplitude acoustics drops Q (shock losses).

the resonator as a mechanical RLC: mass = acoustic inertance $\rho L/S$, compliance = $V/(\rho c^2)$, resistance = radiation + viscous: $Z_{\text{in}}$ of Oly 03 is the acoustic RLC impedance — identical mathematics.

### spherical spreading and the radiating surface




### resonance curve: width f1/Q




REMEMBER / Q = omega E / P_loss ties ring-down, bandwidth and gain; small sources are poor radiators.

# Sign-free Doppler: one vector formula for every case

Vector Doppler f' = f(c − v_o·r)/(c − v_s·r), projections replace cases.

## Sign-free Doppler: one vector formula for every case

Let $\hat{\mathbf{r}}$ point from source to observer at the emission event; let $\mathbf{v}_s, \mathbf{v}_o$ be the velocities of source and observer relative to the medium; $c$ the sound speed in the medium. Then the received frequency is $\mathbf{f' = f\frac{c - v_o \cdot \hat{r}}{c - v_s \cdot \hat{r}}}$ — no signs to memorise: projections carry the signs. Check 1D: source approaching ( $\mathbf{v}_s$ along $+\hat{\mathbf{r}}$ toward observer... observer at larger $r$ : $\mathbf{v}_s \cdot \hat{\mathbf{r}} = +u_s$ ... hmm careful: $\hat{\mathbf{r}}$ from source to observer: source moving toward observer means $\mathbf{v}_s$ has positive projection on $\hat{\mathbf{r}}$ : denominator $c - u_s$ ok the approach rise; observer approaching means $\mathbf{v}_o$ points AGAINST $\hat{\mathbf{r}}$ : $\mathbf{v}_o \cdot \hat{\mathbf{r}} = -u_o$ : numerator $c + u_o$ ok exactly JEE-24.

## Sign-free Doppler: one vector formula for every case (cont.)

Wind: replace $c\hat{\mathbf{r}}$ -type bookkeeping by the medium velocity $\mathbf{w}$ : sound velocity in ground frame $\mathbf{c} = c\hat{\mathbf{r}} + \mathbf{w}$ ; the clean invariant form measures $\mathbf{v}_s, \mathbf{v}_o$ RELATIVE TO THE MEDIUM and $c$ in the medium:

$f' = f\frac{c - (\mathbf{v}_o - \mathbf{w}) \cdot \hat{\mathbf{r}}}{c - (\mathbf{v}_s - \mathbf{w}) \cdot \hat{\mathbf{r}}}$ ... the robust recipe:

transform to the medium frame (subtract $\mathbf{w}$ from every ground velocity), apply the box, transform back (frequency is frame-invariant in classical acoustics for the observer present... the RECEIVED frequency is what the observer measures in their own frame: do all algebra in the medium frame then quote f').

## Sign-free Doppler: one vector formula for every case (cont.)

Derivation: in the medium frame the wavefronts have spacing $\lambda' = \frac{c - \mathbf{v}_s \cdot \hat{\mathbf{r}}}{f}$ (source compresses along $\hat{\mathbf{r}}$ ) and the observer sweeps them at rate $\frac{c - \mathbf{v}_o \cdot \hat{\mathbf{r}}}{\lambda'}$ : quotient = the box. Nothing else.

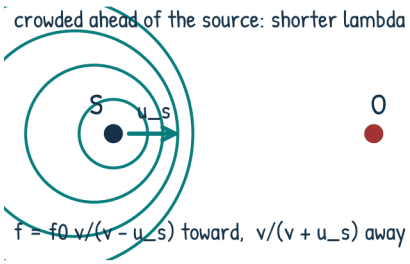
WHY / the dot products are the entire content of the 2D, angled, wind, and echo cases: every hard Doppler problem reduces to drawing $\hat{\mathbf{r}}$ and projecting — the box replaces a case library.

## Sign-free Doppler: one vector formula for every case (cont.)

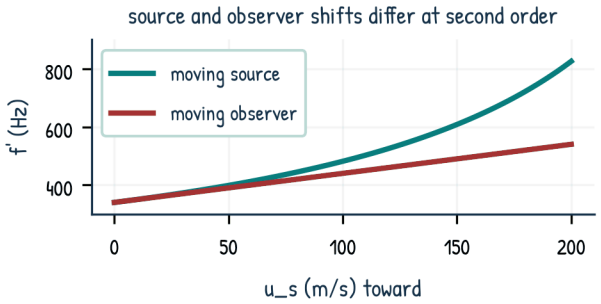
CHECK / specialise all cases: observer at rest: numerator $c$ ; source at rest: denominator $c$ ; both at rest in wind: ratio $(c - \mathbf{w} \cdot \hat{\mathbf{r}})/(c - \mathbf{w} \cdot \hat{\mathbf{r}}) = 1$ ok no shift in uniform wind (JEE-25 convention); tangential source ( $\mathbf{v}_s \perp \hat{\mathbf{r}}$ ): $f' = f$ at that instant ok glide midpoint.

(a) Source on the x-axis at $(vt, b)$ moving +x at $u$ ; observer at origin: $\hat{\mathbf{r}} = \frac{(-vt,\ -b)}{\sqrt{v^2t^2 + b^2}}$ ... set $ut$ for source position: $\hat{\mathbf{r}} = \frac{(-ut,\ -b)}{\sqrt{u^2t^2 + b^2}}$ ;

### the vector formula: project onto the line of sight




### the same glide curve from one formula




REMEMBER / f = f0 (c − v_o.r)/(c − v_s.r): one vector formula replaces the entire case library.

# Sign-free Doppler: one vector formula for every case (cont.)

Vector Doppler f' = f(c − v_o·r)/(c − v_s·r), projections replace cases.

## Vector clinic: worked projections (cont.)

$\mathbf{v}_s \cdot \hat{\mathbf{r}} = \dfrac{-u^2 t}{\sqrt{u^2 t^2 + b^2}} = -u\cos\theta(t)$:

$f' = f\dfrac{c}{c + u\cos\theta}$ ... check sign convention: approaching ($t < 0$): $\cos\theta < 0$: denominator $< c$: $f' > f$ ok matches glide picture (with $\theta$ measured from the velocity direction).

(b) Observer moving at angle: pedestrian walking at 2 m/s at $60^\circ$ to the line from a 800 Hz siren 50 m away, $u_s = 30$ m/s: project both; the pedestrian term is $2\cos 60 = 1$ m/s: $f' = 800\dfrac{343 - 1}{343 - 30\cos\theta}$ – geometry first, numbers last.

## Vector clinic: worked projections (cont.)

(c) Echo as two applications of the box: target = intermediate observer/source; compose: $f'' = f\dfrac{c - \mathbf{v}_T \cdot \hat{\mathbf{r}}}{c - \mathbf{v}_T \cdot (-\hat{\mathbf{r}})}$ -ish for monostatic... yields $\dfrac{c+u}{c-u}$ for approach ok JEE-27.

CHECK / frame sanity: the medium frame is singled out by the physics (sound needs a medium); the formula is NOT Lorentz-invariant – the contrast with light (relativistic Doppler: $f' = f\sqrt{\frac{1-\beta}{1+\beta}}$ approach... $\frac{1}{\gamma(1 \mp \beta)}$ factors) is a favourite exam discussion.

## Where the case library hides inside the box

Case reduction: (i) 1D: projections $\pm u$; (ii) 2D: $u\cos\theta$ instantaneous; (iii) wind: shift to medium frame; (iv) moving reflector: two box applications; (v) accelerating: integrate projections over emission events (Oly 07); (vi) source passing: $\theta(t)$ from the geometry – the glide (Oly 08). The "four cases" of JEE-24 are four projection patterns of one sentence.

Second-order asymmetry explained: the box is exact; expanding $f' = f\dfrac{c + u_o}{c - u_s}$: the $u_s$ denominator is a geometric series while the $u_o$ numerator is finite – the asymmetry is built into the kinematics of wavefront emission vs collection.

## Where the case library hides inside the box (cont.)

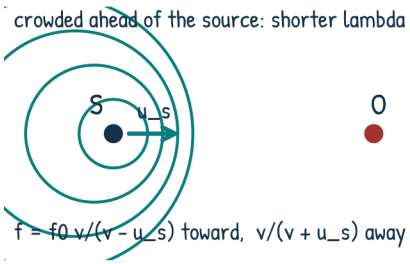
Applications: acoustic Doppler velocimetry (measure $\mathbf{v}_o \cdot \hat{\mathbf{r}}$ of particles), anemometry (wind from $\mathbf{w} \cdot \hat{\mathbf{r}}$ via two beams), bat sonar (dual use of box + echo), fetal Doppler (blood-cell reflectors).

WHY / the vector box makes reasoning CHECKABLE: after any case analysis, re-derive with projections; agreement validates the sign work – the antidote to sign errors.

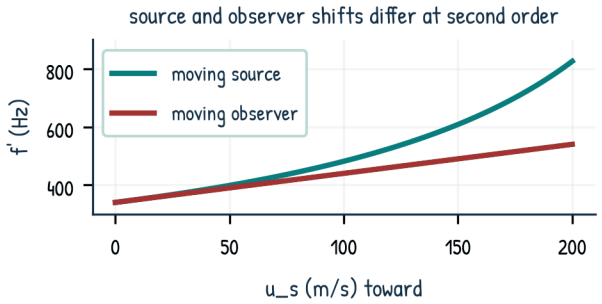
(a) $f = 500$ Hz source at $u_s = 40$ m/s at $30^\circ$ to the line of sight, receding side: $f' = 500\dfrac{343}{343 + 40\cos 30^\circ} = 500 \cdot \dfrac{343}{377.6} = 454$ Hz.

(b) Both moving toward each other at $u = 20$ each, 1D: $f' = 500 \cdot \dfrac{363}{323} = 562$ Hz.

### the vector formula: project onto the line of sight




### the same glide curve from one formula




REMEMBER / f = f0 (c − v_o.r)/(c − v_s.r): one vector formula replaces the entire case library.

# Sign-free Doppler: one vector formula for every case (cont.)

Vector Doppler f' = f(c − v_o·r)/(c − v_s·r), projections replace cases.

## Box drills (answers in one line each) (cont.)

(c) Wind 15 m/s along the line, both at rest in ground: $f' = f$ (no shift), travel time downwind $L/(c+w)$ ok both parts of JEE-25 from one formula.

Reverse use: measure $f'$ from two positions to get the velocity projections; two beams at known angles solve for the 2D velocity vector — the Doppler ultrasound vector flow meter.

relativistic check: for light in vacuum the medium-frame concept dies; the invariant formula is $f' = f\gamma(1 - \beta\cos\theta)$ — same projection structure, different dynamics: a satisfying comparison answer.

## Accelerated motion: integrating the instantaneous shift

For a source with arbitrary velocity $\mathbf{v}_s(t)$ the observer hears $f'(t_e) = f\dfrac{c}{c - \mathbf{v}_s(t_e)\cdot\hat{\mathbf{r}}(t_e)}$ evaluated at the EMISSION time $t_e$: what is heard at time $t$ was emitted at $t_e = t - R(t_e)/c$ (implicit!) — retarded-time bookkeeping. Constant velocity: closed form (JEE-26 glide); constant acceleration along the line: $f'(t_e) = f\dfrac{c}{c - at_e}$: frequency diverges as the source approaches speed $c$ ... for $at_e \to c$; reached at finite time: sonic boom geometry (JEE-28).

## Accelerated motion: integrating the instantaneous shift (cont.)

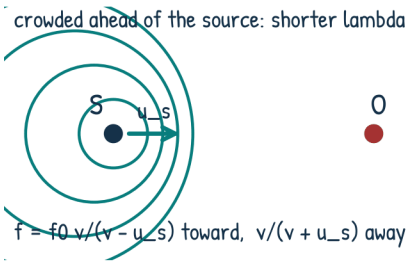
The glide-slope example as an integral: closest approach at $t = 0$, speed $u$, impact $b$: $f'(t) = f\dfrac{c}{c - u^2 t/\sqrt{b^2 + u^2 t^2}}$ (Oly 06 projections): linearising near $t = 0$: $f' \approx f\left(1 + \dfrac{u^2 t}{cb}\right)$: slope $\dfrac{df'}{dt}|_0 = \dfrac{fu^2}{cb}$ downward after the pass ok the JEE-26 formula as the first term of the exact expansion.

## Accelerated motion: integrating the instantaneous shift (cont.)

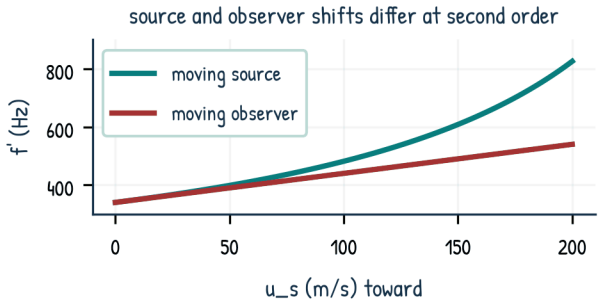
Slowly accelerating radial motion: quasi-static shift $f'(t) \approx f(1 + v_r(t)/c)$: the frequency curve is the RADIAL VELOCITY curve scaled by $f/c$ — acoustic velocimetry: measuring $f'(t)$ reads $v_r(t)$ directly (differencing successive spectra).

WHY / the retarded-time structure is why accelerating-source problems are implicit: the emission point itself moves; solvable cases are (i) slow variation (quasi-static), (ii) constant velocity (closed form), (iii) special geometries (circular motion: Bessel sidebands).

## the vector formula: project onto the line of sight




## the same glide curve from one formula




REMEMBER / f = f0 (c − v_o.r)/(c − v_s.r): one vector formula replaces the entire case library.

# Accelerated motion: integrating the instantaneous shift

Glide from the vector box, slope fu²/(cb), retarded-time thinking.

## Accelerated motion: integrating the instantaneous shift (cont.)

CHECK / circular motion check: source circling at radius $R$, angular rate $\Omega$: $v_r = u\cos(\Omega t)$ -ish: $f' = f\left(1 + \frac{u}{c}\cos\Omega t\right)$ first order: PURE frequency modulation with sidebands $f \pm \Omega/2\pi$ — the ambulance-on-a-roundabout FM signal.

Setup recap: straight path, impact $b$, speed $u$, stationary observer, emission continuous. $\cos\theta(t) = \frac{ut}{\sqrt{b^2 + u^2t^2}}$ (sign per convention); $f'(t) = f\frac{c}{c - u\cos\theta(t)}$; values: $t = 0$: $f' = f$; $t \to \pm\infty$:

## Glide geometry: the exact curve and its slope (cont.)

$f\frac{c}{c \mp u}$; slope at 0: $\mp\frac{fu^2}{cb}$ (decreasing through the pass).

Characteristic width: the glide occupies $\Delta t \sim b/u$ (the time to swing $\theta$ by a radian).

Spectrogram picture: a hyperbolic-like chirp symmetric in time but NOT in frequency (approach side higher by the $c/(c-u)$ vs $c/(c+u)$ asymmetry); a bat or porpoise hears the whole swoop and uses the slope for ranging (Oly 08).

Numbers (corrected source example): $f = 1000$ Hz, $u = 20$ m/s, $b = 100$ m, $c = 340$: slope 11.8 Hz/s; $f'(\pm 5$ s$) = 1043.2/960.1$ Hz; asymmetry at equal times: $83.1$ Hz total swing: the integral of the slope over the pass.

## Glide geometry: the exact curve and its slope (cont.)

CHECK / consistency of slope and swing: average slope over $\pm 5$ s $\approx 83/10 = 8.3$ Hz/s vs peak 11.8 — smooth rolloff ok; far-field asymptotes reproduce the 1D pair 1043.5/960.5 Hz within 0.5 Hz at $\pm 5$ s... hmm far asymptotes are $1000 \cdot 340/320 = 1062.5$ and $1000 \cdot 340/360 = 944.4$: the pass values are closer to $f$ than the asymptotes ok correct shape.

(a) A car accelerating from rest toward a listener at $a = 2$ m/s²: $v_r = at$: $f' = f(1 + at/c)$ -ish until significant: at $t = 10$ s, $v = 20$ m/s: $f' = 1.06f$: the pitch RISES continuously (unlike constant-velocity pass which glides down after). (b) A dropped source: $v_r = gt$ toward a listener below: same linear chirp downward in observed... source falling AWAY from an observer above:

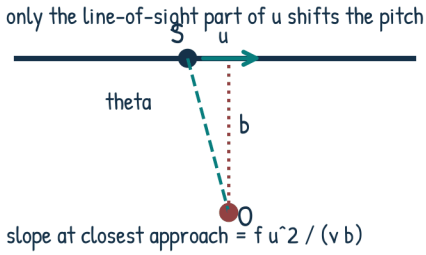
## Accelerated-source problem patterns (cont.)

$f' = f(1 - gt/c)$: a falling whistle chirps down at $gf/c = 28.7$ Hz/s for $f = 1000$.

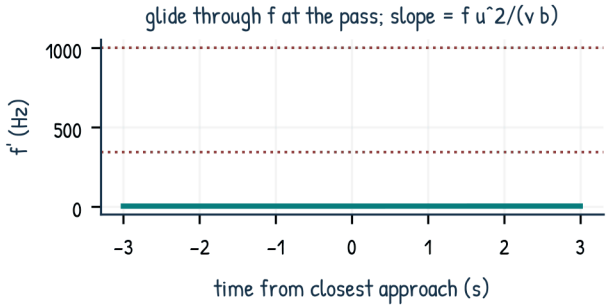
Retarded-time example: source at distance $d$ moving away at $u$ emitted at $t = 0$: received at $t = d/c$; the NEXT wavefront is received at $t = (d + u/f)/c$ -ish: spacing stretched by $(c+u)/c$ ok the receding factor derived from scratch.

rotating and vibrating sources: turbine blades, sirens, ultrasound transducers: the spectrum is a set of sidebands $f \pm nf_{\text{rot}}$ — Bessel-function amplitudes; the exam-accessible part is the first-order FM picture.

### closest approach: radial speed crosses zero




### the glide: slope f u^2 / (v b)




REMEMBER / Accelerated motion: integrate the instantaneous shift at emission time; the glide is the first lesson.

# Accelerated motion: integrating the instantaneous shift (cont.)

Glide from the vector box, slope fu²/(cb), retarded-time thinking.

## Slope-and-swing drills

(a) $f = 600$ Hz, $u = 25$ m/s, $b = 150$ m: slope $= 600 \cdot 625/(340 \cdot 150) = 7.35$ Hz/s. (b) "The observed pitch of a passing cycle horn (500 Hz) falls 60 Hz within 3 s of the pass: estimate $u^2/b$": slope 20 Hz/s $= fu^2/(cb)$: $u^2/b = 20 \cdot 340/500 = 13.6$ m/s$^2$-ish units: with $u = 25$: $b = 46$ m. (c) Bat closing on an insect: both glide AND radial approach: total slope = glide term + $fa/c$ from acceleration. What can go wrong: near-field ( $b \lesssim \lambda$ ) wave geometry breaks the ray picture; supersonic $u > c$ invalidates the subsonic formula entirely (JEE-28); turbulent air adds scintillation (amplitude noise), not frequency shift.

## Slope-and-swing drills (cont.)

WHY / the glide is the acoustic fingerprint of the geometry: slope gives $u^2/b$ (one equation, two unknowns — need the swing width $b/u$ or a second measurement to separate $u$ and $b$ — the structure of exam problems.

If the emission frequency $f$ and the source speed $u$ are known, one second of spectrogram near closest approach gives $b = \dfrac{fu^2}{c|slope|}$ (from Oly 07). If instead $b$ is known (e.g. flyby altitude), the slope gives $u$:

## Ranging from frequency slope: the acoustic McAfee method (cont.)

$u = \sqrt{\dfrac{c\,b\,|slope|}{f}}$. With BOTH slope and swing width $\Delta t \sim b/u$: the pair determines $u$ and $b$ independently: $u = |slope|\, c\, \Delta t/f$ -ish... precisely combine $S = fu^2/(cb)$ and $W = b/u$: $u = \sqrt{ScW/f}$ ... derive: $SW = fu^2/(cb) \cdot b/u = fu/c$:

$u = \dfrac{SWc}{f}$ ok and $b = uW$.

## Ranging from frequency slope: the acoustic McAfee method (cont.)

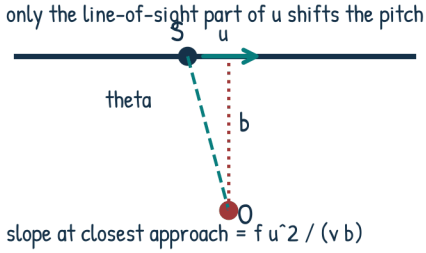
Echo version (radar/sonar McAfee): measure the beat $\Delta f = 2fu/c$ between direct and echo (JEE-27):

$u = \dfrac{\Delta f\, c}{2f}$ directly; combine with the echo delay $\tau = 2R/c$ for range; the pair ( $\Delta f$ , $\tau$ ) gives ( $u$ , $R$ ) of the target — the full tracking state at one instant.

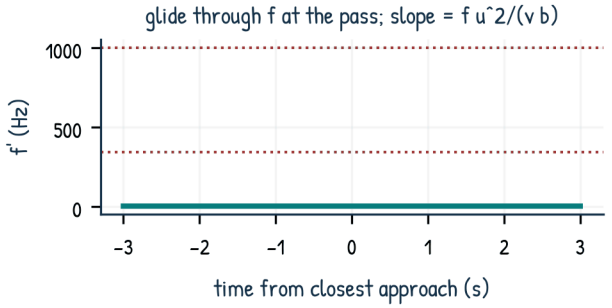
Accuracy: frequency slope measurable to ~1% from a clean spectrogram: $b$ to ~2%; the method needs no timing sync — pure frequency metrology: robust to amplitude noise; fails when $u$ is unknown along with $b$ (add a second pass or beam).

WHY / the slope method converts a TIME-domain difficult problem (nanosecond timing) into a FREQUENCY-domain easy one (Hz over seconds) — the deep reason Doppler ranging works.

### closest approach: radial speed crosses zero




### the glide: slope f u^2 / (v b)




REMEMBER / Accelerated motion: integrate the instantaneous shift at emission time; the glide is the first lesson.

# Ranging from frequency slope: the acoustic McAfee method

Ranging via slope u²/b, echo beats, two-beam tracking, method table.

## Ranging from frequency slope: the acoustic McAfee method (cont.)

CHECK / units: $[fu^2/(c\,slope)]$ = Hz m $^2$ s $^{-2}$ / (m s $^{-1}$ Hz s $^{-1}$) = m ok; numbers: $f = 1000$, $u = 20$, $c = 340$, slope 11.8: $b = 100$ m ok recovers the geometry.

Two beams at angles $\theta_1, \theta_2$ to the target line: measure radial speeds $u\cos\theta_i$ ... solve: $u = \dfrac{v_{r1} - v_{r2}}{\cos\theta_1 - \cos\theta_2}$ -style elimination; three beams give the full 3D velocity vector — Doppler sodar and ultrasound vector flow imaging.

Track differentiation: successive ($u$, $R$) estimates give acceleration $a = du/dt$; the glide trajectory then fits ($b$, $u$, $a$) — the standard manoeuvring-target model.

## Two-beam and multi-beam Doppler tracking (cont.)

GPS-free ground speed: doppler log on ships (beam to the seabed), air data from pitot + acoustic cross-check; police lidar/radar: EM version of the same math ($c \to 3\times10^8$, double shift always).

CHECK / double-beam example: $\theta = 30°$ beams, measured $v_r = 43.3$ and 0 m/s (one beam along the motion): $u = 50$ m/s ok cross-check $50\cos 30 = 43.3$ ok.

(a) "A jet of known speed 250 m/s at unknown altitude passes; the 400 Hz tone glides at 300 Hz/s through the closest point: find altitude": $b = fu^2/(cS) = 400\cdot62500/(340\cdot300) = 245$ m — absurdly low for a jet: sanity-check rejects the numbers (maybe $S$ read wrong from the spectrogram) — the discipline of orders of magnitude.

## Slope-method problem patterns (cont.)

(b) "A drone hovers at 40 m emitting 2 kHz; a cyclist approaches at 8 m/s: beat with the echo": $\Delta f = 2fu_r/c$ with radial 8 m/s: 94 Hz beats — drone speed sensors use exactly this (acoustic flow sensors). (c) Bat 1: measure slope of the insect pass at $b = 2$ m, $u = 4$ m/s, 40 kHz: slope $= 40000\cdot16/(340\cdot2) = 941$ Hz/s — bats chirp-sweep over ~2 kHz in ms, resolvable.

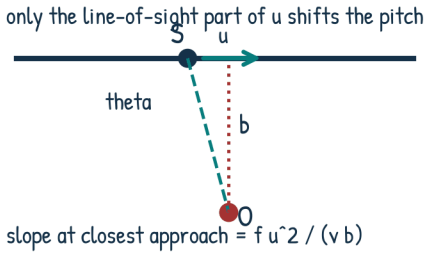
WHY / combining slope + swing + echo delay separates $u$, $b$, $R$ — the exam trinity of ranging unknowns and their three equations.

## Method comparison table (which tool for which question)

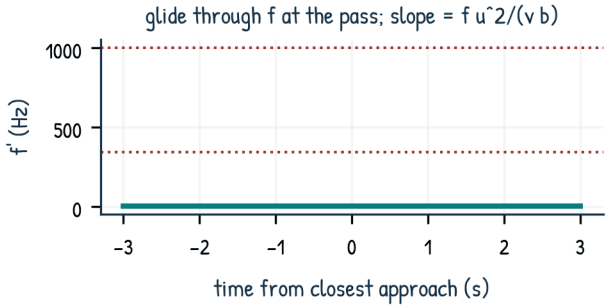
Echo delay: gives RANGE $R = c\tau/2$ directly; needs timing. Doppler beat: gives RADIAL SPEED $\Delta f = 2fu/c$; needs clean spectrum. Glide slope: gives $u^2/b$; needs a pass geometry. Echo delay + beat: state $(u, R)$ ok. Slope + swing: state $(u, b)$ ok. Two beams: vector velocity ok. Time-of-flight + speed of sound thermometer: $c$ from $T$ (JEE-08).

WATCH / Error budget: frequency measurements are precise ($10^{-4}$ easily) but $c$ depends on $T$ ($\pm0.2$ % per K) and wind ($\pm w/c$): for precision ranging, measure $T$ and $w$ or use two-way symmetric methods that cancel wind to first order (JEE-25).

### slope plus swing gives u and b separately




### echo beat: the target speed in one number




REMEMBER / Frequency slope gives u^2/b; echo beats give u; two measurements give the full state.

# Ranging from frequency slope: the acoustic McAfee method (cont.)

Ranging via slope u²/b, echo beats, two-beam tracking, method table.

## Method comparison table (which tool for which question) (cont.)

the same toolkit works from bat sonar to medical Doppler to police radar; changing $c$ and $f$ scales the numbers, never the structure — the closing argument for learning the vector box.

For a duct of slowly varying cross-section $A(x)$ with negligible reflection (smooth taper, $|dA/dx| \ll kA$), conservation of time-averaged acoustic power gives $I(x)A(x) =$ const. Since $I = \dfrac{\Delta P_0^2}{2Z}$ and $I = \frac{1}{2} Z \omega^2 s_0^2$ in the same medium:

pressure amplitude $\Delta P_0 \propto A^{-1/2}$, displacement amplitude $s_0 \propto A^{-1/2}$, intensity $I \propto A^{-1}$ — a horn concentrates SOUND PRESSURE as $1/\sqrt{A}$ from throat to mouth.

## Slowly varying ducts and horns: I(x) A(x) = constant (WKB) (cont.)

WKB validity: the reflection coefficient per taper step is $R \sim \left(\dfrac{d \ln Z_c}{2k\,dx}\right)^2$ -ish with $Z_c = \rho c / A$ the local characteristic impedance... practical rule: $R$ negligible when the area changes by much less than a factor 2 per wavelength; abrupt changes reflect (JEE-16 boundary theory).

Exponential horn: $A(x) = A_0 e^{mx}$: the horn equation has wave solutions with local wavenumber $k_{\text{eff}} = \sqrt{k^2 - (m/2)^2}$: below cutoff $k < m/2$: EVANESCENT (horn stops passing — the horn cutoff frequency); the flare constant $m$ sets the cutoff $f_c = \dfrac{mc}{4\pi}$ -ish... standard: $f_c = \dfrac{mc}{4\pi}$ for exponential:

## Slowly varying ducts and horns: I(x) A(x) = constant (WKB) (cont.)

low-frequency horns must flare gently (large mouth) — why trumpets end in big bells.

WHY / power conservation is the entire WKB content: the wave equation in a slowly varying duct with the ansatz $\Delta P = \mathcal{A}(x) e^{i \int k dx}$ gives $\mathcal{A}\sqrt{Z_c}$ -type invariants; the exam physics is the $A(x)I(x) =$ const line.

CHECK / convergence example: throat 1 cm² to mouth 100 cm²: $I$ drops 100x (20 dB) but $\Delta P_0$ RISES 10x: horns trade intensity for pressure at the throat — exactly what a driver needs (impedance matching, Oly 01: $Z_{\text{in}} \sim Z/A$ -ish).

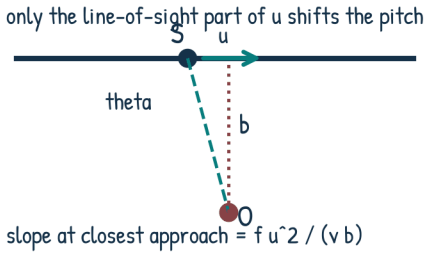
## Horns as impedance transformers (why they work)

A driver of area $S_d$ couples to air poorly (Oly 05 radiation resistance); a horn raises the throat loading: $Z_{\text{throat}} \approx \rho c / A_{\text{mouth}}$ (geometric gain): the mouth impedance $\rho c$ scaled by area ratio: efficiency up to ~50% vs a few % for a bare cone. The same mathematics as the middle ear (Oly 01) and the megaphone of antiquity.

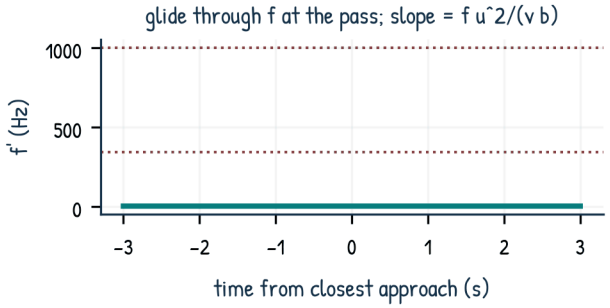
Cutoff and directivity: below $f_c$ the horn is a poor radiator (evanescent throat); above, the mouth radiates a beam of angle $\theta \sim \sin^{-1}(\lambda/(\pi d))$ -ish for mouth size $d$: megaphones beam high frequencies — the shouter hears their own voice differently than listeners do.

Mufflers and ducts: the reverse design — expansions and side branches reflect (Oly 03 traps); a smooth diffuser recovers pressure (wind tunnels, exhausts).

## slope plus swing gives u and b separately




## echo beat: the target speed in one number




REMEMBER / Frequency slope gives u^2/b; echo beats give u; two measurements give the full state.

# Slowly varying ducts and horns: I(x) A(x) = constant (WKB)

WKB duct I·A = const, amplitude $\propto$ A^(−1/2), horn cutoff and matching.

## Horns as impedance transformers (why they work) (cont.)

CHECK / numbers: exponential horn $m = 5\ \mathrm{m}^{-1}$: $f_c = 5 \cdot 343/(4\pi) = 136$ Hz: audible bass passes; $m = 50$: $f_c = 1.36$ kHz: a trumpet-ish cutoff ok plausible instrument numbers.

(a) A duct halves its area: $I$ doubles, $\Delta P_0$ rises $\sqrt{2}$: the speed of the air in the throat rises as $u \propto I/\Delta P \propto 1/\sqrt{A}$ ... $u = \Delta P/Z$ stays? $\Delta P_0 \propto A^{-1/2}$, $u_0 = \Delta P_0/Z \propto A^{-1/2}$: velocity rises as the duct narrows ok (like a river through a constriction).

## WKB drill set (cont.)

(b) "Design a horn to halve the pressure amplitude of a 1 Pa source at the mouth": expand area 4x.

Horn equation quick form: $\frac{1}{A}\frac{d}{dx}\left(A\frac{dP}{dx}\right) + k^2 P = 0$:

WKB solution $P \propto \frac{e^{\pm ikx}}{\sqrt{A}}$ ok recovering the amplitude law; exact for cones and exponentials in terms of Bessel/Hankel functions.

shallow-water waves obey the same $IA =$ const rule (energy conservation); the "water hose narrowing" and tidal-bore funnelling are WKB in geophysics.

## When WKB fails (scope guards)

(i) Area jumps: full reflection analysis (Oly 02/03). (ii) $A \to 0$ (a closed end): WKB diverges; the boundary layer solution takes over (JEE-17). (iii) $dA/dx \sim kA$: significant reflection; (iv) losses (viscous boundary layers: $\alpha \propto \sqrt{f}/a$ in narrow tubes) break power conservation: horns must stay wide vs the viscous skin depth.

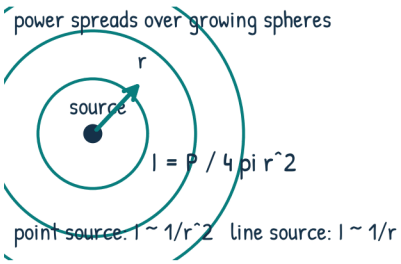
Helmholtz resonator limit: when the duct is shorter than $\lambda/4$, the lumped-parameter RLC model replaces WKB entirely — asymptotic methods need their validity conditions stated (the exam phrase: "slowly varying compared with $\lambda$").

## Elasticity of solids: three speeds from three moduli

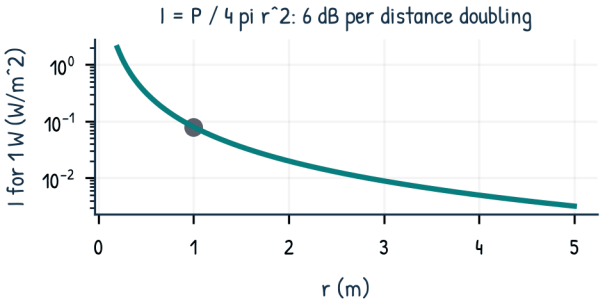
WHY / knowing where an approximation dies is what separates a formula user from a physicist: every WKB answer should carry the check $|d\ln A/dx| \ll k$.

Longitudinal rod (free sides): $v_{\mathrm{rod}} = \sqrt{Y/\rho}$; bulk longitudinal (infinite solid): $v_P = \sqrt{(K + 4G/3)/\rho}$; bulk transverse (shear): $v_S = \sqrt{G/\rho}$. Relations: $Y = 2G(1+\nu) = 3K(1-2\nu)$; with $\nu = 0.29$ (steel): $v_P/v_S = \sqrt{\frac{2(1-\nu)}{1-2\nu}} = 1.75$; $v_P/v_{\mathrm{rod}} = \sqrt{\frac{(1-\nu)}{(1+\nu)(1-2\nu)}}$ -ish ≈ 1.15–1.2. Seismology: P-waves = $v_P$, S-waves = $v_S$; S cannot propagate in fluids ($G = 0$) — the evidence Earth outer core is liquid.

### slowly varying duct: power I A = constant




### intensity falls as area grows




REMEMBER / Slowly varying ducts carry I A = constant: amplitudes fall as A^(−1/2); horns match impedance.

# Elasticity of solids: three speeds from three moduli

Rod/bulk/shear speeds, Lamé derivation sketch, seismology patterns.

## Elasticity of solids: three speeds from three moduli (cont.)

Derivation sketch: Newton 2nd law + Hooke tensor law $\sigma_{ij} = \lambda\,\mathrm{tr}(\epsilon)\delta_{ij} + 2G\epsilon_{ij}$ (Lamé $\lambda$, $G$): plane longitudinal wave: $\rho\ddot{u} = (\lambda + 2G)\nabla^2 u$: $v_P = \sqrt{(\lambda + 2G)/\rho} = \sqrt{(K + 4G/3)/\rho}$ since $K = \lambda + 2G/3$; transverse: $\rho\ddot{u}_\perp = G\nabla^2 u_\perp$: $v_S = \sqrt{G/\rho}$ ok.

Numbers: steel ($K = 160$, $G = 79$ GPa, $\rho = 7800$): $v_P = 5830$, $v_S = 3180$, $v_{\mathrm{rod}} = 5060$ m/s. Granite: $v_P \approx 5200$, $v_S \approx 3000$. Glass ($G = 30$ GPa): $v_S = 2130$. Rubber ($G \approx 1$ MPa, $\rho = 1100$): $v_S = 30$ m/s — slow and lossy.

## Elasticity of solids: three speeds from three moduli (cont.)

WHY / three combinations of two elastic constants + density: the rod speed is a boundary-conditioned mixture; the bulk pair is the intrinsic content of the elasticity tensor; Poisson ratio interpolates.

CHECK / inequalities: $v_P > v_{\mathrm{rod}} > v_S$ for $0 < \nu < 0.5$ ok (steel: 5830 > 5060 > 3180); $\nu \to 0$: $v_P = v_{\mathrm{rod}}$; $\nu \to 0.5$: $v_P \to \infty$ (incompressible: $K \to \infty$).

Thin rod free-free (or clamped at centre): longitudinal modes $f_n = n\dfrac{v_{\mathrm{rod}}}{2L}$; clamped-free: $f_n = (2n-1)\dfrac{v_{\mathrm{rod}}}{4L}$ (odd, like the closed pipe); the same ladders as air columns with the solid speed. Dispersive corrections: for rods of finite thickness ($ka \gtrsim 1$) Pochhammer-Chree modes appear: the simple $\sqrt{Y/\rho}$ is the long-wave limit.

## Waves in rods: which boundary, which ladder (cont.)

Kundt tube (JEE-21) is the rod-speed measurement in reverse: $v_{\mathrm{rod}}$ known gives $v_{\mathrm{gas}}$; with a gas of known speed the rod speed comes out: the two-medium frequency match $f = v_{\mathrm{rod}}/\lambda_{\mathrm{rod}} = v_{\mathrm{gas}}/\lambda_{\mathrm{gas}}$.

Damping: metals ring (low internal friction, Q ~ $10^3$ – $10^4$ for rods); lead and rubber damp (high loss tangent): struck rods of different metals have characteristic ring times — an exam qualitative question.

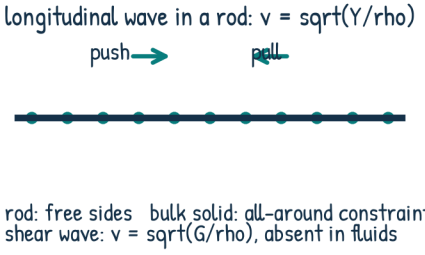
CHECK / steel rod 1 m clamped mid: $f = 5060/2 = 2530$ Hz; in Kundt with dust heaps 6.5 cm apart: $v_{\mathrm{gas}} = 5060 \times 0.065 = 329$ m/s (cold air ok plausible lab).

## Solid-state acoustic problems (exam patterns)

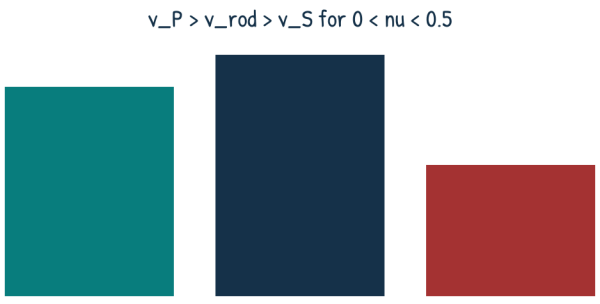
(a) "A longitudinal wave travels a steel rail; a transverse follows: arrival split over 2 km": $\Delta t = 2000(1/3180 - 1/5830)$ ... wait P arrives first: $\Delta t = 2000(1/3180 - 1/5830) = 0.285$ s ok the seismograph method for locating epicentres. (b) "Find $\nu$ from the ratio $v_P/v_S = 1.8$": $\frac{2(1-\nu)}{1-2\nu} = 3.24$: $\nu = 0.27$.

(c) "A rod and a string of the same length and fundamental: compare $v$": equal $f$: $v_{\mathrm{rod}}/2L = \sqrt{T/\mu}/2L$: the speed equality sets the tension.

rod, bulk and shear waves in one bar




three elastic speeds side by side




REMEMBER / Three elastic speeds: rod sqrt(Y/rho), P sqrt((K+4G/3)/rho), S sqrt(G/rho); fluids carry no shear.

# Elasticity of solids: three speeds from three moduli (cont.)

Rod/bulk/shear speeds, Lamé derivation sketch, seismology patterns.

## Solid-state acoustic problems (exam patterns) (cont.)

Composite rods: series springs: effective compliance adds: $1/Y_{\text{eff}} = \sum L_i/(Y_i L)$ -ish: wave speed in layered rods uses transfer matrices (outside syllabus; the 1D spring model is in).

dispersion in real solids: $v(\omega)$ bends near lattice resonances (Brillouin); ultrasound velocity metrology uses $v_P$, $v_S$ to get $K$, $G$, $\nu$ non-destructively — the engineering reading of this card.

## Numbers drill (three speeds)

Aluminium ( $Y = 70$ GPa, $G = 26$, $\rho = 2700$ ): $v_{\text{rod}} = 5100$, $v_P = 6270$, $v_S = 3100$ m/s. Copper ( $Y = 120$, $G = 45$, $\rho = 8900$ ): rod 3670, $v_P = 4760$, $v_S = 2250$. Granite ( $K = 50$ GPa, $G = 25$ GPa, $\rho = 2700$ ): $v_P = 5250$, $v_S = 3040$. Water ( $B = 2.2$ GPa): $v = 1483$, no shear.

Seismic locate: P-S time difference $\Delta t = d(1/v_S - 1/v_P)$: $d = \dfrac{\Delta t}{1/v_S - 1/v_P}$: for $\Delta t = 10$ s in granite: $d = 10/(3.29 - 1.90) \times 10^{-4}$ ... units: $d = \Delta t/(1/3040 - 1/5250) = 10/(3.29 \times 10^{-4} - 1.90 \times 10^{-4}) = 72$ km ok standard method.

## Numbers drill (three speeds) (cont.)

WHY / the pair (arrival times of P and S) plus one speed model locates events in three dimensions — the seismology version of echo + beat (Oly 08): two measurements, two unknowns.

Full supersonic toolkit: cone half-angle $\sin\alpha = 1/M$; observer at height $h$ below the flight line: boom delay after closest pass $t = \dfrac{h\sqrt{M^2-1}}{Mv}$ (envelope derivation of JEE-28); emission point of the heard boom at $t_e = -t/(M^2-1)$ (BEFORE the pass);

the jet position at hearing is $x = h\sqrt{M^2-1}$ past the closest point. The three quantities answer three different questions — always identify which is asked.

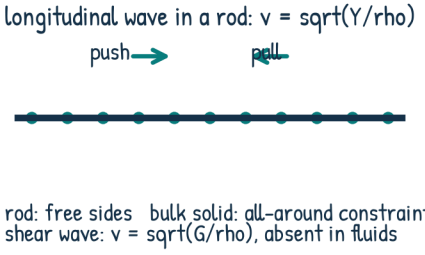
## Mach geometry clinic: cones, delays, and emission points (cont.)

Boom carpet: the shock surface intersects the ground along a swath of width $w = h\cot\alpha \cdot 2$ -ish for the full cone crossing... the ground trace: at any instant the cone-ground intersection is a line moving at $Mv$; observers under the flight path receive the boom once; those off the track receive it at slant geometry; the overpressure decays slowly with altitude ( $\propto h^{-3/4}$ -ish empirically).

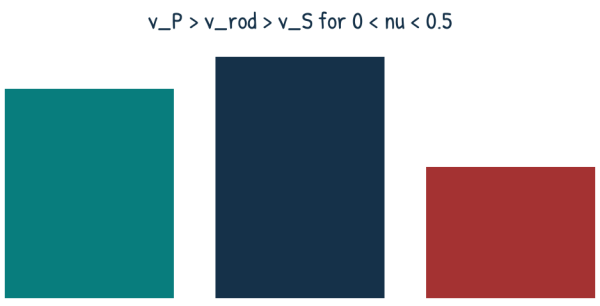
N-wave structure: the boom is a compression-rarefaction doublet of duration $\tau \sim \dfrac{\ell}{v}\cdot$ (aircraft length factor):

Concorde: 0.1–0.3 s at 100–200 Pa; the spectrum of an N-wave is flat to $f \sim 1/\tau$ then falls: characteristic boom "crack".

### rod, bulk and shear waves in one bar




### three elastic speeds side by side




REMEMBER / Three elastic speeds: rod sqrt(Y/rho), P sqrt((K+4G/3)/rho), S sqrt(G/rho); fluids carry no shear.

# Mach geometry clinic: cones, delays, and emission points

Mach clinic: delay formula, emission point, boom carpet, drills.

## Mach geometry clinic: cones, delays, and emission points (cont.)

WHY / the clinic separates three confusable distances: emission distance (from $t_e$), hearing-time position ($h\cot\alpha$), and slant range ($h/\sin\alpha$): the source paper Q13 conflates the first two (SOURCE-AUDIT).

CHECK / self-consistency: $t = h\cot\alpha/u_s$ (cone sweep) $= h\sqrt{M^2-1}/(Mv)$ (envelope) ok identical expressions; $M=1.2$, $h=10$ km: 16.4 s; $M=3$: 27.7 s — the non-monotonic delay again ok JEE-28 grid.

## Angles and triangles: the clinic worksheet

Step 1: compute $c(T)$ (JEE-08). Step 2: $M = u/c$; $\alpha = \sin^{-1}(1/M)$. Step 3: draw the cone BEHIND the source; the observer is inside after the sweep. Step 4: identify the asked quantity: delay ($h\cot\alpha/u$), emission geometry ($t_e = -t/(M^2-1)$), or slant ($h/\sin\alpha$). Step 5: check limits ($M=1$: degenerate; $M\to\infty$: delay $\to h/v$).

Two observers: different delays from different $h$ and lateral offsets: the difference in arrival times LOCATES the flight line — inverse boom mapping (used to track supersonic tests).

Sonic boom from manoeuvres: turns pile shocks (focused superbooms); the clinic covers straight-and-level only.

## Angles and triangles: the clinic worksheet (cont.)

CHECK / work the clinic on source Q13: $M=2$, $h=10$ km, $v=340$: $\alpha=30^\circ$ ok; $t=25.5$ s ok; $t_e=-8.5$ s ok; jet at hearing 17.32 km past ok — all four numbers consistent with the one diagram.

(a) $M=1.5$, $h=5$ km: $\alpha=41.8^\circ$, $t = 5000\sqrt{1.25}/(1.5\cdot340) = 11.0$ s... $\sqrt{M^2-1}=1.118$: $t = 5000\cdot1.118/510 = 10.96$ s ok. (b) "Observers at 10 and 20 km lateral offset hear the boom 8 s apart: find $M$ and altitude": two equations ($t_i = \sqrt{h^2+x_i^2}\cdot$... solve geometry: classic two-station problem. (c) "How long after the FLASH does the boom arrive for $M=2$, $h=8$ km?": flash $\approx$ overhead instant: delay $= 20.4$ s.

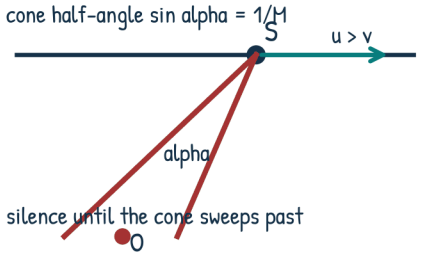
## Clinic drills (cont.)

WATCH / Error sensitivities: $\delta t/t = \delta h/h + \delta M\cdot($ $M/(M^2-1) - 1/M$ )-ish: at $M=2$: $\delta t/t \approx \delta h/h + 0.25\delta M/M$: Mach errors matter at low supersonic.

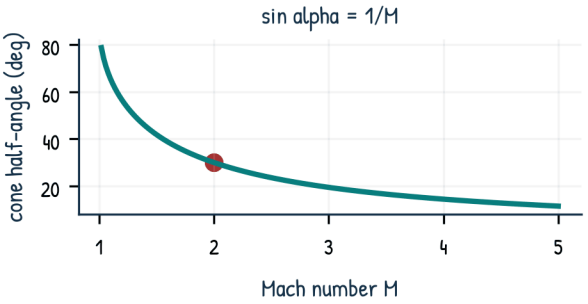
focused booms from turns can double overpressure (Whitham theory); "superboom" events over Oklahoma City (1960s tests) seeded the modern low-boom research (X-59) — history with the clinic math.

Mach number $M = u/c$; Mach angle $\alpha = \sin^{-1}(1/M)$; N-wave: two-shock signature; overpressure $\Delta P$: 50–500 Pa typical; carpet: ground swath; boom delay: closest-approach to hearing; superboom: focused enhancement; cutoff Mach: $M=1$ exactly (infinite linear pile-up).

boom delay geometry from the cone




delay vs Mach: peaks near M = sqrt(2)




REMEMBER / Boom geometry clinic: delay h sqrt(M^2-1)/(M v), emission before the pass, carpet on the ground.

# Mach geometry clinic: cones, delays, and emission points (cont.)

Mach clinic: delay formula, emission point, boom carpet, drills.

## Sonic-boom terminology card (cont.)

Regulation: supersonic flight over land banned for civil (US 1973 FAA rule) because of boom annoyance; Concorde flew supersonic only over water; X-59 targets 75 Pa "thump" to reopen the question.

WHY / precise vocabulary earns the method marks: examiners look for "cone", "shock", "delay from closest approach", and the $\sin\alpha = 1/M$ relation in any boom answer.

Kit formula set (air unless stated): $c = 331.3\sqrt{T/273}$ m/s; $Z = \rho c \approx 412$ rayl at 20 C; $\lambda = c/f$; $\Delta P_0 = Z\omega s_0$; $I = \Delta P_0^2/(2Z)$; $\beta = 10\log(I/10^{-12})$ dB; point source $I = P/(4\pi r^2)$; beats $|f_1 - f_2|$; Doppler $\frac{c+u_o}{c-u_s}$; echo $\frac{c+u}{c-u}$; pipes $v/4L$, $v/2L$; $e = 0.6r$; $Q = \omega\mathcal{E}/P_{\text{loss}}$.

## Acoustic estimation kit: building your own numbers (cont.)

Sizing tricks: $\lambda = 343/f$: 1 kHz $\leftrightarrow$ 34 cm; 100 Hz $\leftrightarrow$ 3.4 m; 10 kHz $\leftrightarrow$ 3.4 cm. $s_0$ at threshold (1 kHz): 0.01 nm; at pain: 10 $\mu$ m. Time: sound 1 km in 2.9 s; 1 mile in 4.7 s; light-sound delay 1 km: 2.9 s (the "count the seconds" method).

WHY / the kit is the chapter in twelve lines: every worked example of the source is a substitution; every Olympiad problem is two or three kit lines chained — the closing consolidation of the set.

CHECK / re-derive the source Q3 chain from the kit: 0.1 W isotropic at 100 m: $I = 8\times10^{-5}$ W/m$^2$; $\beta = 79$ dB; $\Delta P_0 = 0.26$ Pa; $s_0 = 0.3\,\mu$m at 340 Hz ok all five kit lines agree with the worked answer.

## Acoustic estimation kit: building your own numbers (cont.)

With these, estimate any sound problem to ~10% in one line each.

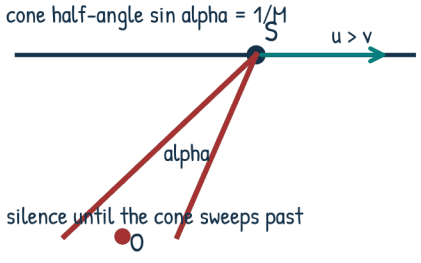
Order-of-magnitude anchors: speech 60 dB at 1 m (~ $10\,\mu$ W acoustic); shout 1 mW; rock concert front 0.1–1 W/m$^2$; threshold of hearing $10^{-12}$ W/m$^2$ = 20 $\mu$ Pa; pain 20 Pa; room modes $c/2L$ (~170 Hz for a 1 m scale); ear canal 3.4 kHz; vocal formants 300/2300 Hz; jet 130 dB at 30 m; thunder 120 dB at 100 m.

## Estimate-first discipline (the Fermi habit)

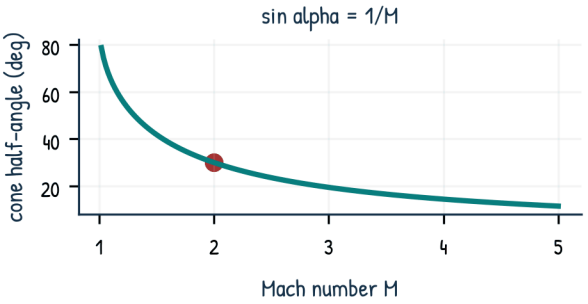
Before writing equations: estimate the answer in rounded numbers (factor-of-3 accuracy); then compute exactly; the two must agree within the rounded uncertainty — if not, a unit or a factor of $2\pi$ is wrong. Worked: "level 2 m from a 0.5 W siren": estimate $I \sim 0.5/50 \sim 0.01$ W/m$^2$: $\beta \sim 100$ dB; exact: 92 dB — same ballpark; the estimate catches the "$P/4\pi r^2$ vs $P/r^2$" family of slips immediately.

Unit traps the kit prevents: dB without a reference; $\lambda$ in cm vs m in $v = f\lambda$; temperature in Celsius inside $\sqrt{T}$; $M$ in g/mol inside $RT/M$; intensity vs pressure amplitude factors of 2; peak vs RMS $\sqrt{2}$.

### boom delay geometry from the cone




### delay vs Mach: peaks near M = sqrt(2)




REMEMBER / Boom geometry clinic: delay h sqrt(M^2-1)/(M v), emission before the pass, carpet on the ground.

# Acoustic estimation kit: building your own numbers

Estimation kit, Fermi drills, chapter map and mark-loss list.

## Estimate-first discipline (the Fermi habit) (cont.)

Reality checks: sound speed in gas hundreds, liquid ~1500, solid thousands of m/s; pressures $\Delta P \ll P_{\text{atm}}$; displacement amplitudes nm-mm; if a computed $s_0$ is 5 cm for speech, it is wrong.

WHY / examiners hide unit slips inside otherwise correct derivations; the estimate-first habit detects them in seconds — the meta-skill this kit trains.

(a) "How far can a shout be heard in still air?" assume 1 mW and 40 dB threshold at the ear: $I = 10^{-8}$ W/m$^2$: $r = \sqrt{P/(4\pi I)} = 2.8$ km geometric; with 0.1 dB/m absorption: $\beta = 30 - 20\log r - 0.1r = 40$ ... solve: r ~ 150 m realistic — geometry overestimates, absorption rules. (b) "Pitch of a 20 cm closed pipe at 30 C": $c = 349$: $f = c/4L = 436$ Hz (order 400).

## Fermi drill set (estimate, then compute) (cont.)

(c) "Beat between a 440 Hz fork and a source at 15 m/s toward": $\Delta f = fu/c = 19$ Hz (order 20).

Chain estimate: "speed of a train from its horn": hear 470 Hz approaching, 400 receding... $u = c(f_+ - f_-)/(f_+ + f_-) = 343 \times 70/870 = 27.6$ m/s (order 100 km/h ok 99 km/h) — estimate 30 m/s ok.

Fermi problems for the set: estimate the acoustic power of a piano (1 W), the sound energy of a conversation in an hour ($10^{-2}$ J), the pressure of a sonic boom at 10 km (order 100 Pa) — all kit lines.

## The kit as a closing map (chapter index)

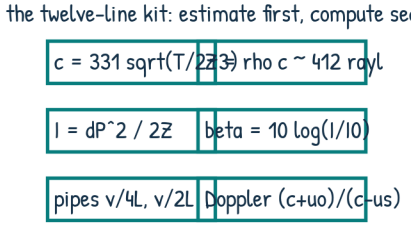
Wave basics (SN-01..04): $s$, $\Delta P = -B\partial_x s$, $v = \sqrt{B/\rho}$. Media (SN-05..09, OLY-10): $\sqrt{\gamma RT/M}$, $\sqrt{Y/\rho}$, $\sqrt{(K + 4G/3)/\rho}$. Energy (SN-10..12, OLY-05): $I = \Delta P_0^2/2Z$, dB. Interference (SN-13..16, OLY-04): superposition, phasors, standing waves. Pipes (SN-17..21, OLY-03/09): ladders, end correction, impedance, WKB. Motion (SN-22..28, OLY-06..08/11): Doppler box, echo, glide, Mach. Boundaries (OLY-01/02): $Z$, $R$. Estimation (OLY-12): the kit.

## The kit as a closing map (chapter index) (cont.)

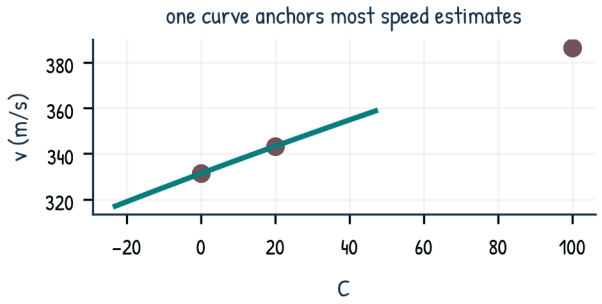
Where marks are lost (the short list): kelvin vs Celsius; $e$ sign in the resonance tube ($l = \lambda/4 - e$); overtone counting; Doppler sign conventions; peak vs RMS; intensity vs amplitude dB factors; factor 2 in Quincke ($\delta = 2\Delta L$); dust heaps at $\lambda/2$.

WHY / the kit card is the revision sheet: if any line is unfamiliar, that sheet of the set deserves another pass — the teacher-style close of the chapter.

### the twelve-line estimation kit




### one speed curve anchors the estimates




REMEMBER / The kit: estimate first with rounded numbers, compute second, demand agreement within the estimate.